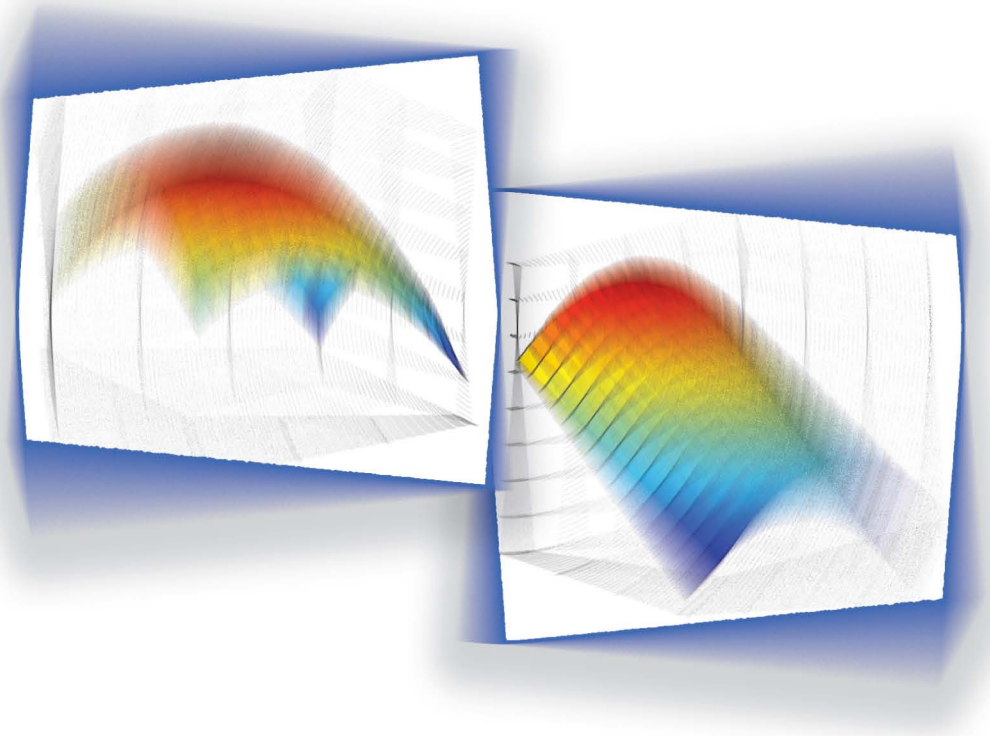
RANDOM PHENOMENA

FUNDAMENTALS OF PROBABILITY AND STATISTICS FOR ENGINEERS



BABATUNDE A. OGUNNAIKE

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Babatunde A. Ogunnaike

Random Phenomena

Fundamentals of Probability & Statistics for Engineers

I frame no hypothesis; for whatever is not deduced from the phenomenon is to be called a hypothesis; and hypotheses, whether metaphysical or physical, whether of occult qualities or mechanical, have no place in experimental philosophy.

Sir Isaac Newton (1642–1727)

In Memoriam

In acknowledgment of the debt of birth I can never repay, I humbly dedicate this book to the memory of my father, my mother, and my statistics mentor at Wisconsin.

Adesijibomi Ogundẹrọ Ogunnaike
1922–2002
Some men fly as eagles, free
But few with grace to the same degree
as when you rise upward to fly
to avoid sparrows in a crowded sky

Ayọọla Oluronkẹ Ogunnaike
1931–2005
Some who only search for silver and gold
Soon find what they cannot hold;
You searched after God's own heart,
and left behind, too soon, your pilgrim's chart

William Gordon Hunter
1937–1986
See what ripples great teachers make
With but one inspiring finger
Touching, once, the young mind's lake

Preface

In an age characterized by the democratization of quantification, where data about every conceivable phenomenon is available somewhere and easily accessible from just about anywhere, it is becoming just as important that the "educated" person *also* be conversant with how to handle data, and be able to understand what the data say as well as what they don't say. Of course, this has always been true of scientists and engineers—individuals whose profession requires them to be involved in the acquisition, analysis, interpretation and exploitation of data in one form or another; but it is even more so now. Engineers now work in non-traditional areas ranging from molecular biology to finance; physicists work with material scientists and economists; and the problems to be solved continue to widen in scope, becoming more interdisciplinary as the traditional disciplinary boundaries disappear altogether or are being restructured.

In writing this book, I have been particularly cognizant of these basic facts of 21st century science and engineering. And yet while most scientists and engineers are well-trained in problem formulation and problem solving when *all* the entities involved are considered deterministic in character, many remain uncomfortable with problems involving random variations, if such problems cannot be idealized and reduced to the more familiar "deterministic" types. Even after going through the usual one-semester course in "Engineering Statistics," the discomfort persists. Of all the reasons for this circumstance, the most compelling is this: most of these students tend to perceive their training in statistics more as a set of instructions on *what* to do and *how* to do it, than as a training in fundamental principles of random phenomena. Such students are then uncomfortable when they encounter problems that are not quite similar to those covered in class; they lack the fundamentals to attack new and unfamiliar problems. The purpose of this book is to address this issue directly by presenting basic fundamental principles, methods, and tools for formulating and solving engineering problems that involve randomly varying phenomena. The premise is that by emphasizing fundamentals and basic principles, and then illustrating these with examples, the reader will be better equipped to deal with a range of problems wider than that explicitly covered in the book. This important point is expanded further in the prelude chapter, Chapter 0.

Scope and Organization

Developing a textbook that will achieve the objective stated above poses the usual challenge of balancing breadth and depth—an optimization problem with no unique solution. But there is also the additional constraint that the curriculum in most programs can usually only accommodate a one-semester course in engineering statistics—if they can find space for it at all. As all teachers of this material know, finding a universally acceptable compromise solution is impossible. What this text offers is enough material for a two-semester introductory sequence in probability and statistics for scientists and engineers, and with it, the flexibility of several options for using the material. We envisage the following three categories, for which more detailed recommendations for coverage will be provided shortly:

- Category I: The two-semester undergraduate sequence.
- Category II: The traditional one-semester undergraduate course.
- Category III: The one-semester beginning graduate course.

The material has been tested and refined over more than a decade, in the classroom (at the University of Delaware; at the African University of Science and Technology (AUST), in Abuja, Nigeria; at the African Institute of Mathematical Sciences (AIMS) in Muizenberg, South Africa), and in short courses presented to industrial participants at DuPont, W. L. Gore, SIEMENS, the Food and Drug Administration (FDA), and many others through the University of Delaware's Engineering Outreach program.

The book is organized into 5 parts, after a brief prelude in Chapter 0 where the book's organizing principles are expounded. Part I (Chapters 1 and 2) provides foundational material for understanding the fundamental nature of random variability. Part II (Chapters 3–7) focuses on probability. Chapter 3 introduces the fundamentals of probability theory, and Chapters 4 and 5 extend these to the concept of the random variable and its distribution, for the single and the multidimensional random variable. Chapter 6 is devoted to random variable transformations, and Chapter 7 contains the first of a trilogy of case studies, this one devoted to two problems with substantial historical significance.

Part III (Chapters 8–11) is devoted entirely to developing and analyzing probability models for specific random variables. The distinguishing characteristics of the presentation in Chapters 8 and 9, respectively for discrete and continuous random variables, is that each model is developed from underlying phenomenological mechanisms. Chapter 10 introduces the idea of information and entropy as an alternative means of determining appropriate probability models when only partial knowledge is available about the random variable in question. Chapter 11 presents the second case study, on in-vitro fertilization (IVF), as an application of probability models. The chapter illustrates the development, validation, and use of probability modeling on a contemporary problem with significant practical implications.

The core of statistics is presented in Part IV (Chapters 12–20). Chapter 12 lays the foundation with an introduction to the concepts and ideas behind statistics, before the coverage begins in earnest in Chapter 13 with sampling theory, continuing with statistical inference, estimation and hypothesis testing, in Chapters 14 and 15, and regression analysis in Chapter 16. Chapter 17 introduces the important but oft-neglected issue of probability model validation, while Chapter 18 on nonparametric methods extends the ideas of Chapters 14 and 15 to those cases where the usual probability model assumptions (mostly the normality assumption) are invalid. Chapter 19 presents an overview treatment of design of experiments. The third and final set of case studies is presented in Chapter 20 to illustrate the application of various aspects of statistics to real-life problems.

Part V (Chapters 21–23) showcases the "application" of probability and statistics with a hand-selected set of "special topics": reliability and life testing in Chapter 21, quality assurance and control in Chapter 22, and multivariate analysis in Chapter 23. Each has roots in probability and statistics, but all have evolved into *bona fide* subject matters in their own rights.

Key Features

Before presenting suggestions of how to cover the material for various audiences, I think it is important to point out some of the key features of the textbook.

1. Approach. This book takes a more fundamental, "first-principles" approach to the issue of dealing with random variability and uncertainty in engineering problems. As a result, for example, the treatment of probability distributions for random variables (Chapters 8–10) is based on a derivation of each model from phenomenological mechanisms, allowing the reader to see the subterraneous roots from which these probability models sprang. The reader is then able to see, for instance, how the Poisson model arises either as a limiting case of the binomial random variable, or from the phenomenon of observing in finite-sized intervals of time or space, rare events with low probabilities of occurrence; or how the Gaussian model arises from an accumulation of small random perturbations.

2. Examples and Case Studies. This fundamental approach noted above is integrated with practical applications in the form of a generous amount of examples but also with the inclusion of three chapter-length application case studies, one each for probability, probability distributions, and statistics. In addition to the usual traditional staples, many of the in-chapter examples have been drawn from non-traditional applications in molecular biology (e.g., DNA replication origin distributions; gene expression data, etc.), from finance and business, and from population demographics.

3. Computers, Computer Software, On-line Resources. As expanded further in the Appendix, the availability of computers has transformed the teaching and learning of probability and statistics. Statistical software packages are now so widely available that many of what used to be staples of traditional probability and statistics textbooks—tricks for carrying out various computations, approximation techniques, and especially printed statistical tables—are now essentially obsolete. All the examples in this book were carried out with MINITAB, and I fully expect each student and instructor to have access to one such statistical package. In this book, therefore, we depart from tradition and do *not* include any statistical tables. Instead, we have included in the Appendix a compilation of useful information about some popular software packages, on-line electronic versions of statistical tables, and a few other on-line resources such as on-line electronic statistics handbooks, and websites with data sets.

4. Questions, Exercises, Application Problems, Projects. No one feels truly confident about a subject matter without having tackled (and solved!) some problems; and a useful textbook ought to provide a good selection that offers a broad range of challenges. Here is what is available in this book:

- *Review Questions*: are short, specific questions designed to test the reader's basic comprehension at the end of each chapter (with the exception of the chapters on case studies: Chapters 7, 11 and 20). If you can answer all the review questions at the end of each chapter, you know and understand the material; if not, revisit the relevant portion and rectify the revealed deficiency.

- *Exercises*: are "technical" problems designed to provide the opportunity to master the mechanics behind a single concept. Some may therefore be purely "mechanical" in the sense of requiring basic computations; some may require filling in the steps deliberately "left as an exercise to the reader;" some may have the flavor of an application; but the focus is usually a single aspect of a topic covered in the text, or a straightforward extension thereof.

- *Application Problems*: are more substantial practical problems whose solutions usually require integrating various concepts (some obvious, some not) and deploying the appropriate set of tools. Many of these are drawn from the literature and involve real applications and actual data sets. In such cases, the references are provided, and the reader may wish to consult some of them for additional background and perspective, if necessary.

- *Project Assignments*: allow deeper exploration of a few selected issues covered in a chapter, mostly as a way of extending the coverage and also to provide opportunities for creativity. By definition, these involve a significant amount of work and also require report-writing. This book

offers a total of nine such projects, in Chapters 11, 20, 22, and 23. They are a good way for students to learn how to plan, design, and execute projects and to develop writing and reporting skills. (Each graduate student that has taken the CHEG 604 and CHEG 867 courses at the University of Delaware has had to do a term project of this type.)

5. Data Sets. All the data sets used in each chapter, whether in the chapter itself, in an example, or in the exercises or application problems, are made available on-line and on CD, which can be found at www.routledge.com/9781420044973.

Suggested Coverage

Of the three categories mentioned earlier, a methodical coverage of the entire textbook is only possible for Category I, in a two-semester undergraduate sequence. For this group, the following is one possible approach to dividing the material up into instruction modules for each semester:

- **First Semester**

  Module 1 (Foundations): Chapters 0–2.

  Module 2 (Probability): Chapters 3, 4, 5, and 7.

  Module 3 (Probability Models): Chapter 8 (omit detailed derivations and Section 8.7.2), Chapter 9 (omit detailed derivations), and Chapter 11 (cover Sections 11.4 and 11.5 selectively; omit Section 11.6).

  Module 4 (Introduction to Statistics/Sampling): Chapters 12 and 13.

  Module 5 (Statistical Inference): Chapter 14 (omit Section 14.6), Chapter 15 (omit Sections 15.8 and 15.9), Chapter 16 (omit Sections 16.4.3, 16.4.4, and 16.5.2), and Chapter 17.

  Module 6 (Design of Experiments): Chapter 19 (cover Sections 19.3–19.4 lightly; omit Section 19.10) and Chapter 20.

- **Second Semester**

  Module 7 (Probability and Models): Chapter 6 (with *ad hoc* reference to Chapters 4 and 5), Chapters 8 and 9 (include details omitted in the first semester), and Chapter 10.

  Module 8 (Statistical Inference): Chapter 14 (Bayesian estimation, Section 14.6), Chapter 15 (Sections 15.8 and 15.9), Chapter 16 (Sections 16.4.3, 16.4.4, and 16.5.2), and Chapter 18.

  Module 9 (Applications): Select one of Chapter 21, 22, or 23. (For chemical engineers, and anyone planning to work in the manufacturing industry, I recommend Chapter 22.)

With this as a basic template, other variations can be designed as appropriate.

For example, those who can only afford one semester (Category II) may adopt the first semester suggestion above, to which *I recommend adding Chapter 22 at the end.*

The beginning graduate one-semester course (Cateogory III) may also be based on the first semester suggestion above, but with the following additional recommendations: (i) cover of all the recommended chapters *fully*; (ii) add Chapter 23 on multivariate analysis; and (iii) in lieu of a final examination, assign at least one, possibly two, of the nine projects.

This will make for a hectic semester, but graduate students should be able to handle the workload.

A second, perhaps more straightforward, recommendation for a two-semester sequence is to devote the first semester to Probability (Chapters 0–11), and the second to Statistics (Chapters 12–20) along with one of the three application chapters (Chapters 21, 22, and 23).

Acknowledgments

Pulling off a project of this magnitude requires the support and generous assistance of many colleagues, students, and family. Their genuine words of encouragement and the occasional (innocent and not-so-innocent) inquiry about the status of "the book" all contributed to making sure that this potentially endless project was actually finished. At the risk of leaving someone out, I feel some deserve particular mention. I begin with, in alphabetical order, Marc Birtwistle, Ketan Detroja, Claudio Gelmi (Chile), Mary McDonald, Vinay Prasad (Alberta, Canada), Paul Taylor (AIMS, Muizenberg, South Africa), and Carissa Young. These are colleagues, former and current students, and postdocs, who patiently waded through many versions of various chapters, offered invaluable comments and caught many of the manuscript errors, typographical and otherwise. It is a safe bet that the manuscript still contains a random number of these errors (few and Poisson distributed, I hope!) but whatever errors remain are my responsibility. I encourage readers to let me know of the ones they find.

I wish to thank my University of Delaware colleagues, Antony Beris and especially Dion Vlachos, with whom I often shared the responsibility of teaching CHEG 867 to beginning graduate students. Their insight into what the statistics component of the course should contain was invaluable (as were the occasional Greek lessons!). Of my other colleagues, I want to thank Dennis Williams of LyondellBasell, for his interest and comments, and then single out former fellow "DuPonters" Mike Piovoso, whose fingerprint is recognizable on the illustrative example of Chapter 23, Rafi Sela, now a Six-Sigma Master Black Belt, Mike Deaton of James Madison University, and Ron Pearson, whose near-encyclopedic knowledge never ceases to amaze me. Many of the ideas, problems, and approaches evident in this book arose from those discussions and collaborations from many years ago. Of my other academic colleagues, I wish to thank Carl Laird of Texas A & M for reading some of

the chapters, Joe Qin of USC for various suggestions, and Jim Rawlings of Wisconsin with whom I have carried on a long-running discussion about probability and estimation because of his own interests and expertise in this area. David Bacon and John MacGregor, pioneers in the application of statistics and probability in chemical engineering, deserve my thanks for their early encouragement about the project and for providing the occasional commentary. I also wish to take this opportunity to acknowledge the influence and encouragement of my chemical engineering mentor, Harmon Ray. I learned more from Harmon than he probably knew he was teaching me. Much of what is in this book carries an echo of his voice and reflects the Wisconsin tradition.

I must not forget my gracious hosts at the École Polytechnique Fédérale de Lausanne (EPFL), Professor Dominique Bonvin (*Merci pour tout, mon ami*) and Professor Vassily Hatzimanikatis (*Ευχαριστω πολυ παλιοφιλε*: "Efharisto poli paliofile"). Without their generous hospitality during the months from February through July 2009, it is very likely that this project would have dragged on for far longer. I am also grateful to Michael Amrhein of the Laboratoire d'Automatique at EPFL, and his graduate student, Paman Gujral, who both took time to review several chapters and provided additional useful references for Chapter 23. My thanks go to Allison Shatkin and Marsha Pronin of CRC Press/Taylor and Francis for their professionalism in guiding the project through the various phases of the editorial process all the way to production.

And now to family. Many thanks are due to my sons, Damini and Deji, who have had cause to use statistics at various stages of their (still on-going) education: each read and commented on a selected set of chapters. My youngest son, Makinde, still too young to be a proofreader, was nevertheless solicitous of my progress, especially toward the end. More importantly, however, just by "showing up" when he did, and *how*, he confirmed to me without meaning to, that he is a natural-born Bayesian. Finally, the debt of thanks I owe to my wife, Anna, is difficult to express in a few words of prose. She proofread many of the chapter exercises and problems with an incredible eye, and a sensitive ear for the language. But more than that, she knows well what it means to be a "book widow"; without her forbearance, encouragement, and patience, this project would never have been completed.

Babatunde A. Ogunnaike
Newark, Delaware
Lausanne, Switzerland

June 2009

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Chapter 0

Prelude

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Rem tene; verba sequentur.
(Grasp the subject; the words will follow.)

Cato the Elder (234–149 BC)

From weather forecasts and life insurance premiums for non-smokers to clinical tests of experimental drugs and defect rates in manufacturing facilities, and in numerous other ways, *randomly varying phenomena* exert a subtle but pervasive influence on everyday life. In most cases, one can be blissfully ignorant of the true implications of the presence of such phenomena without consequence. In science and engineering, however, the influence of randomly varying phenomena can be such that even apparently simple problems can become dramatically complicated by the presence of random variability—demanding special methods and analysis tools for obtaining valid and useful solutions.

The primary aim of this book is to provide the reader with the basic fundamental principles, methods, and tools for formulating and solving engineering problems involving randomly varying phenomena.

Since this aim can be achieved in several different ways, this chapter is devoted to presenting this book's approach philosophy.

0.1 Approach Philosophy

Engineers are typically well-trained in the "art" of problem formulation and problem solving when *all* the entities involved are considered *deterministic* in character. However, many problems of practical importance involve

randomly varying phenomena of one sort or another; and the vast majority of such problems cannot always be idealized and reduced to the more familiar "deterministic" types without destroying the very essence of the problem. For example, in determining which of 2 catalysts—A or B—provides the greater yield in a chemical manufacturing process, it is well-known that the respective yields Y_A and Y_B, *as observed experimentally*, are randomly varying quantities. Chapter 1 presents a full-scale discussion of this problem. For now, we simply note that with catalyst A, 50 different experiments performed under essentially identical conditions will result in 50 different values (*realizations*) for Y_A. Similarly for catalyst B, one obtains fifty distinct values for Y_B from 50 different experiments replicated under identical conditions. The first 10 experimental data points for this example are shown in the table below.

Y_A %	Y_B %
74.04	75.75
75.29	68.41
75.62	74.19
75.91	68.10
77.21	68.10
75.07	69.23
74.23	70.14
74.92	69.22
76.57	74.17
77.77	70.23

Observe that because of the variability inherent in the data, some of the Y_A values are greater than some of the Y_B values; but the converse is also true—some Y_B values are greater than some Y_A values. So how does one determine—reliably and confidently—which catalyst (if any) really provides the greater yield? Clearly, special methods and analysis tools are required for handling this apparently simple problem: the deterministic idealization of comparing a *single* observed value of Y_A (say the first entry, 74.04) with a corresponding *single* observed value of Y_B (in this case 75.75) is incapable of producing a valid answer. The primary essence of this problem is the variability inherent in the data which masks the fact that one catalyst does in fact provide the greater yield.

This book takes a more fundamental, "first-principles" approach to the issue of dealing with random variability and uncertainty in engineering problems. This is in contrast to the typical "engineering statistics" approach on the one hand, or the "problem-specific" approach on the other. With the former approach, most of the emphasis is on *how* to use certain popular statistical techniques to solve *some* of the most commonly encountered engineering problems, with little or no discussion of *why* the techniques are effective. With the latter approach, a particular topic (say "design of experiments") is selected and dealt with in depth, and the appropriate statistical tools are presented and discussed within the context of the specific problem at the core of the

selected topic. By definition, such an approach excludes all other topics that may be of practical interest, opting to make up in depth what it gives up in breadth.

> *The approach taken in this book is based on the premise that emphasizing fundamentals and basic principles, and then illustrating these with examples, equips the reader with the means of dealing with a range of problems wider than that explicitly covered in the book.*

This approach philosophy is based on the four basic principles discussed next.

0.2 Four Basic Principles

1. *If characterized properly, random phenomena are subject to rigorous mathematical analysis in much the same manner as deterministic phenomena.*

Random phenomena are so-called because they show no apparent regularity, appearing to occur haphazardly—totally at random; the observed variations do not seem to obey any discernible rational laws and therefore appear to be entirely unpredictable. However, the unpredictable irregularities of the individual observations (or, more generally, the "detail") of random phenomena in fact co-exist with surprisingly predictable ensemble, or aggregate, behavior. This fact makes rigorous analysis possible; it also provides the basis for employing the concept and calculus of "probability" to develop a systematic framework for characterizing random phenomena in terms of "probability distribution functions."

The first order of business is therefore to seek to understand random phenomena and to develop techniques for characterizing them appropriately. Part I, titled *FOUNDATIONS: Understanding Random Variability*, and Part II, titled *PROBABILITY: Characterizing Random Variability*, are devoted to these respective tasks. Ultimately, probability—and the probability distribution function—are introduced as the theoretical constructs for efficiently describing *our knowledge* of the real-world phenomena in question.

2. *By focusing on the underlying phenomenological mechanisms, it is possible to develop appropriate theoretical characterizations of random phenomena in terms of ideal models of the observed variability.*

Within the probabilistic framework, the ensemble, or aggregate behavior

of the random phenomenon in question is characterized by its "probability distribution function." In much the same way that theoretical mathematical models are derived from "first-principles" for deterministic phenomena, it is also possible to derive these theoretical probability distribution functions as ideal models that describe our knowledge of the underlying random phenomena. Part III, titled *DISTRIBUTIONS: Modeling Random Variability*, is devoted to the important tasks of developing and analyzing ideal probability models for many random phenomena of practical interest. The end result is a collection of probability distribution functions each derived directly from—and hence explicitly linked to—the underlying random phenomenological mechanisms.

3. *The ensemble (or aggregate) characterization provided by ideal probability models can be used successfully to develop the theoretical basis for solving real problems where one is always limited to dealing with an incomplete collection of individual observations—never the entire aggregate.*

A key defining characteristic of random phenomena is that specific outcomes or observations cannot be predicted with *absolute certainty*. With probabilistic analysis, this otherwise "impossible task" of predicting the unpredictable individual observation or outcome is simply replaced by the analytical task of determining the mathematical *probability* of its occurrence. In many practical problems involving random phenomena, however, there is no avoiding this "impossible task": one is required to deal with, and make decisions about, individual observations, and must therefore confront the inevitable uncertainty that will always be associated with such decisions. *Statistical Theory*, using the aggregate descriptions of probability theory, provides a rational basis not only for making these predictions and decisions about individual observations with confidence, but also for quantifying the degree of uncertainty associated with such decisions.

Part IV, titled *STATISTICS: Quantifying Random Variability*, is devoted to elucidating statistical principles and concepts required for dealing effectively with data as collections of individual observations from random phenomena.

4. *The usefulness and broad applicability of the fundamental principles, analysis methods, and tools provided by probability and statistics are best illustrated with several actual example topics of engineering applications involving random phenomena.*

The manifestations of random phenomena in problems of practical interest are countless, and the range of such problems is itself quite broad: from simple data analysis and experimental designs, to polynomial curve-fitting, and empirical modeling of complex dynamic systems; from quality assurance and control, to state and parameter estimation, and process monitoring and diagnosis, ... etc. The topical headings under which such problems may be organized—design of experiments; regression analysis; time series analysis; etc.—are numerous, and many books have been devoted to each one of them.

Clearly then, the sheer vastness of the subject matter of engineering applications of probability and statistics renders completely unreasonable any hope of comprehensive coverage in a single introductory text.

Nevertheless, how probability and statistics are employed in practice to deal successfully with various problems created by random variability and uncertainty can be discussed in such a way as to equip the student with the tools needed to approach, with confidence, other problems that are not addressed explicitly in this book.

Part V, titled *APPLICATIONS: Dealing with Random Variability in Practice*, consists of three chapters each devoted to a specific application topic of importance in engineering practice. Entire books have been written, and entire courses taught, on each of the topics to which we will devote only one chapter; the coverage is therefore designed to be more illustrative than comprehensive, providing the basis for absorbing and employing more efficiently, the more extensive material presented in these other books or courses.

0.3 Summary and Conclusions

This chapter has been primarily concerned with setting forth this book's approach to presenting the fundamentals and engineering applications of probability and statistics. The four basic principles on which the more fundamental, "first principles" approach is based were presented, providing the rationale for the scope and organization of the material to be presented in the rest of the book.

The approach is designed to produce the following result:

> *A course of study based on this book should provide the reader with a reasonable fundamental understanding of random phenomena, a working knowledge of how to model and analyze such phenomena, and facility with using probability and statistics to cope with random variability and uncertainty in some key engineering problems.*

The book should also prepare the student to absorb and employ the material presented in more problem-specific courses such as design of experiments, time series analysis, regression analysis, statistical process control, etc., a bit more efficiently.

Part I

Foundations

Part I: Foundations

Understanding Random Variability

I shall light a candle of understanding in thine heart which shall not be put out.

Apocrypha: I Esdras 14:25

Part I: Foundations

Understanding Random Variability

- **Chapter 1:** Two Motivating Examples
- **Chapter 2:** Random Phenomena, Variability, and Uncertainty

Chapter 1

Two Motivating Examples

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And coming events cast their shadows before.
(Lochiel's warning.)

Thomas Campbell (1777–1844)

When random variability is genuinely intrinsic to a problem, uncertainty becomes inevitable, but the problem can still be solved systematically and with *confidence*. This—the underlying theme of "Applied Probability and Statistics"—is what this chapter seeks to illustrate with two representative examples. The example problems and the accompanying discussions are intended to serve two main purposes: (i) illustrate the sort of complications caused by the presence of random components in practical problems; and (ii) demonstrate (qualitatively for now) how to solve such problems by formulating them properly and employing appropriate methods and tools. The primary value of this chapter is as a vehicle for placing in context this book's approach to analyzing randomly varying phenomena in engineering and science. It allows us to preview and motivate the key concepts to be developed fully in the remaining chapters.

1.1 Yield Improvement in a Chemical Process

To an engineer or scientist, determining which of two numbers is larger, and by how much, is trivial, in principle requiring no more than the elementary arithmetic operation of subtraction. Identify the two numbers as *individual observations* from two randomly varying quantities, however, and the character of the problem changes significantly: determining—with any *certainty*—which of the random quantities is "larger" and precisely by how much now requires more than mere subtraction. This is the case with our first example.

1.1.1 The Problem

A chemical process using catalyst A (process "A") is being considered as an alternative to the incumbent process using a different catalyst B (process "B"). The decision in favor of one process over the other is to be based on a comparison of the yield Y_A obtained from the challenger, and Y_B from the incumbent, in conjunction with the following economic considerations:

- Achieving target profit objectives for the finished product line requires a process yield *consistently* at or above 74.5%.
- For process A to be a viable alternative, at the barest minimum, its yield must be higher than that for process B.
- Every 1% yield increase over what is currently achievable with the incumbent process translates to significant after tax operating income; however, catalyst A used by the alternative process costs more than catalyst B used by the incumbent process. Including the additional cost of the process modifications required to implement the new technology, *a shift to catalyst A and the new process will be economically viable only if the resulting yield increase exceeds 2%.*

The result of a series of 50 experiments "carefully performed" on each process to determine Y_A and Y_B is shown in Table 1.1. Given only the supplied data, what should the Vice President/General Manager of this business do: authorize a switch to the new process A or stay with the incumbent process?

Mathematical Formulation

Observe that solving this problem requires finding appropriate answers to the following mathematical questions:

1. Is $Y_A \geq 74.5$, and $Y_B \geq 74.5$, consistently?
2. Is $Y_A > Y_B$?

TABLE 1.1: Yield data for process A versus process B

Y_A %		Y_B %	
74.04	75.29	75.75	68.41
75.63	75.92	74.19	68.10
77.21	75.07	68.10	69.23
74.23	74.92	70.14	69.23
76.58	77.77	74.17	70.24
75.05	74.90	70.09	71.91
75.69	75.31	72.63	78.41
75.19	77.93	71.16	73.37
75.37	74.78	70.27	73.64
74.47	72.99	75.82	74.42
73.99	73.32	72.14	78.49
74.90	74.88	74.88	76.33
75.78	79.07	70.89	71.07
75.09	73.87	72.39	72.04
73.88	74.23	74.94	70.02
76.98	74.85	75.64	74.62
75.80	75.22	75.70	67.33
77.53	73.99	72.49	71.71
72.30	76.56	69.98	72.90
77.25	78.31	70.15	70.14
75.06	76.06	74.09	68.78
74.82	75.28	72.91	72.49
76.67	74.39	75.40	76.47
76.79	77.57	69.38	75.47
75.85	77.31	71.37	74.12

3. If yes, is $Y_A - Y_B > 2$?

Clearly, making the proper decision hinges on our ability to answer these questions with confidence.

1.1.2 The Essence of the Problem

Observe that the real essence of the problem is *random variability*: if each experiment had resulted in the same single, constant number for Y_A and another for Y_B, the problem would be "deterministic" in character, and each of the 3 associated questions would be trivial to answer. Instead, the random phenomena inherent in the experimental determination of the "true" process yields have been manifested in the observed variability, so that we are uncertain about the "true" values of Y_A and Y_B, making it not quite as trivial to solve the problem.

The sources of variability in this case can be shown to include the measurement procedure, the measurement device itself, raw materials, and process conditions. The observed variability is therefore intrinsic to the problem and cannot be idealized away. There is no other way to solve this problem rationally without dealing directly with the random variability.

Next, note that Y_A and Y_B data (observations) take on values on a "continuous scale" i.e., yield values are "real" and can be located anywhere on the real line, as opposed to quantities that can take on integer values only (as is the case with the second example discussed later). The variables Y_A and Y_B are therefore said to be "continuous" and this example illustrates decision-making under uncertainty when the random phenomena in question involve "continuous variables."

The main issues with this problem are as follows:

1. *Characterization*: How should the quantities Y_A and Y_B be characterized so that the questions raised above can be answered properly?
2. *Quantification*: Are there such things as "true values" of the quantities Y_A and Y_B? If so, how should these "true values" be best quantified?
3. *Application*: How should the characterization and quantification of Y_A and Y_B be used to answer the 3 questions raised above?

1.1.3 Preliminary "Intuitive" Notions

Before outlining procedures for solving this problem, it is helpful to entertain some notions that the intuition of a good scientist or engineer will suggest. For instance, the concept of the "arithmetic average" of a collection of n data points, $x_1, x_2, x_3, \ldots, x_n$, defined by:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{1.1}$$

is well-known to all scientists and engineers, and the intuitive notion of employing this single computed value to "represent" the data set is almost instinctive. It seems reasonable therefore to consider representing Y_A with the computed average obtained from the data, i.e., $\bar{y}_A = 75.52$, and similarly, representing Y_B with $\bar{y}_B = 72.47$. We may now observe right away that $\bar{y}_A > \bar{y}_B$, which now seems to suggest not only that $Y_A > Y_B$, but since $\bar{y}_A - \bar{y}_B = 3.05$, that the difference in fact exceeds the threshold of 2%.

As intuitively appealing as these arguments might be, they raise some important additional questions:

1. The variability of individual values of the data y_{A_i} around the average value $\bar{y}_A = 75.52$ is noticeable; that of y_{B_i} around the average value $\bar{y}_B = 72.47$ even more so. How confident then are we about the arguments presented above, and in the implied recommendation to prefer process A to B, based as they are on the computed averages? (For example, there are some 8 values of $y_{B_i} > \bar{y}_A$; what should we make of this fact?)

2. Will it (or should it) matter that

$$\begin{aligned} 72.30 < \quad & y_{A_i} < 79.07 \\ 67.33 < \quad & y_{B_i} < 78.49 \end{aligned} \tag{1.2}$$

   so that the observed data are seen to vary over a *range* of yield values that is 11.16 units wide for process B as opposed to 6.77 for A? The averages give no indication of these extents of variability.

3. More fundamentally, is it always a good idea to work with averages? How reasonable is it to characterize the entire data set with the average?

4. If new sets of data are gathered, the new averages computed from them will almost surely differ from the corresponding values computed from the current set of data shown here. Observe therefore that the computed averages $\bar{y}_A$ and $\bar{y}_B$ are themselves clearly subject to random variability. How can we then be sure that using averages offers any advantages, since, like the original data, these averages are also not free from random variability?

5. How were the data themselves collected? What does it mean concretely that the 50 experiments were "carefully performed"? Is it possible that the experimental protocols used may have impaired our ability to answer the questions posed above adequately? Conversely, are there protocols that are particularly calibrated to improve our ability to answer these questions adequately?

Obviously therefore there is a lot more to dealing with this example problem than merely using the intuitively appealing notion of averages.

Let us now consider a second, different but somewhat complementary, example.

TABLE 1.2: Number of "inclusions" on sixty 1-sq-m glass sheets

0	1	1	1	0	0	1	0	2	2
2	0	2	2	3	2	0	0	2	0
1	2	0	1	0	1	0	0	1	1
1	1	5	2	0	0	1	4	1	1
2	1	0	0	1	1	0	0	1	1
1	0	0	2	4	0	1	1	0	1

1.2 Quality Assurance in a Glass Sheet Manufacturing Process

A key measure of product quality in a glass sheet manufacturing process is the optical attribute known as "inclusions"— particulate flaws (of size exceeding 0.5 μm) "included" in an otherwise perfectly clear glass sheet. While it is all but inevitable to find *inclusions* in some products, the best manufacturers produce remarkably few glass sheets with these imperfections; and even then, the actual number of *inclusions* on these imperfect sheets is itself usually very low, perhaps 1 or 2.

The specific example in question involves a manufacturer of 1-sq-m sheets of glass used for various types of building windows. Prior to shipping a batch of manufactured product to customers, a sample of glass sheets from the batch is sent to the company's Quality Control (QC) laboratory where an optical scanning device is used to determine X, the number of *inclusions* in each square-meter sheet. The results for 60 samples from a particular batch is shown in Table 1.2.

This particular set of results caused the supervising QC engineer some concern for the following reasons:

1. Historically, the manufacturing process hardly ever produces sheets with more than 3 *inclusions* per square meter; this batch of 60 has three such sheets: two with 4 *inclusions*, and one with 5.

2. Each 1-sq-m sheet with 3 or fewer *inclusions* is acceptable and can be sold to customers unconditionally; sheets with 4 *inclusions* are marginally acceptable so long as a batch of 1000 sheets does not contain more than 20 such sheets; a sheet with 5 or more inclusions is unacceptable and cannot be shipped to customers. All such sheets found by a customer are sent back to the manufacturer (at the manufacturer's expense) for a full refund. The specific sheet of this type contained in this sample of 60 must therefore be found and eliminated.

3. More importantly, the manufacturing process was designed such that

> when operated properly, there will be no more than 3 unacceptable sheets with 5 or more *inclusions* in each batch of 1000 sheets. The process will be uneconomical otherwise.

The question of interest is this: *Does the QC engineer have a reason to be concerned*? Or, stated mathematically, if X^* is the design value of the number of inclusions per square meter associated with the sheets produced by this process, is there evidence in this sample data that $X > X^*$ so that steps will have to be taken to identify the source of this process performance degradation and then to rectify the problem in order to improve the process performance?

As with the first problem, the primary issue here is also the randomness associated with the variable of interest, X, the number of *inclusions* per square meter of each glass sheet. The value observed for X in the QC lab is a randomly varying quantity, not fixed and deterministic. In this case, however, there is little or no contribution to the observed variability from the measurement device: these particulate *inclusions* are relatively few in number and are easily counted *without error* by the optical device. The variability in raw material characteristics, and in particular the control system's effectiveness in maintaining the process conditions at desired values (in the face of inevitable and unpredictable disturbances to the process) all contribute to whether or not there are imperfections, how many there are per square meter, and where they are located on the sheet. Some sheets come out flawless while others end up with a varying number of *inclusions* that cannot be predicted precisely *à-priori*. Thus, once again, the observed variability must be dealt with directly because it is intrinsic to the problem and cannot be idealized away.

Next, note that the data in Table 1.2, being counts of distinct entities, take on integer values. The variable X is therefore said to be "discrete," so that this example illustrates decision-making when the random phenomena in question involve "discrete variables."

1.3 Outline of a Systematic Approach

Even though the two illustrative problems presented above are different in so many ways (one involves continuous variables, the other a discrete variable; one is concerned with comparing two entities to each other, the other pits a single set of data against a design target), the systematic approach to solving such problems provided by probability and statistics applies to both in a unified way. The fundamental issues at stake may be stated as follows:

> *In light of its defining characteristics of intrinsic variability, how should randomly varying quantities be* characterized *and* quantified precisely *in order to facilitate the solution of practical problems involving them*?

TABLE 1.3: Group classification and frequencies for Y_A data (from the proposed process)

Y_A Group	Frequency	Relative Frequency
71.51-72.50	1	0.02
72.51-73.50	2	0.04
73.51-74.50	9	0.18
74.51-75.50	17	0.34
75.51-76.50	7	0.14
76.51-77.50	8	0.16
77.51-78.50	5	0.10
78.51-79.50	1	0.02
TOTAL	50	1.00

What now follows is a somewhat informal examination of the ideas and concepts behind these time-tested techniques. The purpose is to motivate and provide context for the more formal discussions in upcoming chapters.

1.3.1 Group Classification and Frequency Distributions

Let us revisit the example data sets and consider the following alternative approach to the data representation. Instead of focusing on individual observations as presented in the tables of raw data, what if we sub-divided the observations into small groups (called "bins") and re-organized the raw data in terms of how frequently members of each group occur? One possible result is shown in Tables 1.3 and 1.4 respectively for process A and process B. (A different bin size will lead to a slightly different group classification but the principles remain the same.)

This reclassification indicates, for instance, that for Y_A, there is only one observation between 71.51 and 72.50 (the actual number is 72.30), but there are 17 observations between 74.51 and 75.50; for Y_B on the other hand, 3 observations fall in the [67.51-68.50] group whereas there are 8 observations between 69.51 and 70.50. The "relative frequency" column indicates what *proportion* of the original 50 data points are found in each group. A plot of this reorganization of the data, known as the *histogram*, is shown in Fig 1.1 for Y_A and Fig 1.2 for Y_B.

The histogram, a term first used by Pearson in 1895, is a graphical representation of data from a group-classification and frequency-of-occurrence perspective. Each bar represents a distinct group (or class) within the data set, with the bar height proportional to the group frequency. Because this graphical representation provides a picture of how the data are "distributed" in terms of the frequency of occurrence of each group (how much each group

TABLE 1.4: Group classification and frequencies for Y_B data (from the incumbent process)

Y_B Group	Frequency	Relative Frequency
66.51-67.50	1	0.02
67.51-68.50	3	0.06
68.51-69.50	4	0.08
69.51-70.50	8	0.16
70.51-71.50	4	0.08
71.51-72.50	7	0.14
72.51-73.50	4	0.08
73.51-74.50	6	0.12
74.51-75.50	5	0.10
75.51-76.50	6	0.12
76.51-77.50	0	0.00
77.51-78.50	2	0.04
78.51-79.50	0	0.00
TOTAL	50	1.00

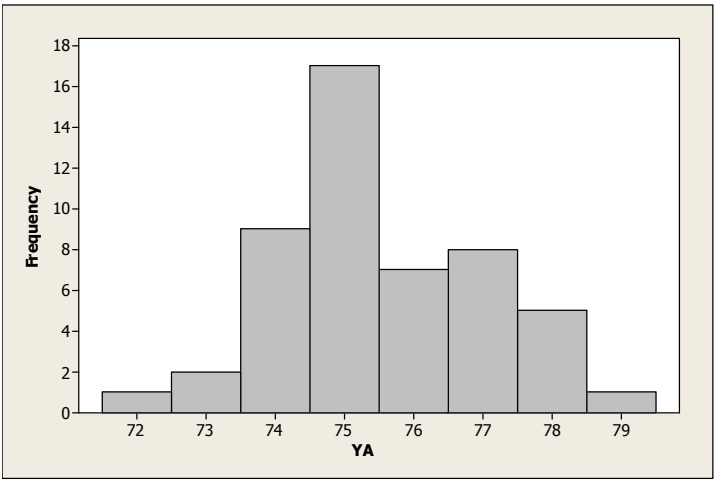


FIGURE 1.1: Histogram for Y_A data.

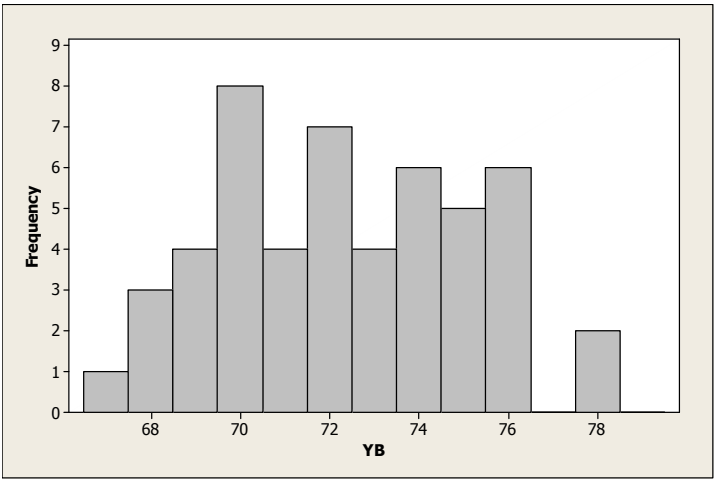


FIGURE 1.2: Histogram for Y_B data.

contributes to the data set), it is often referred to as a *frequency distribution* of the data.

A key advantage of such a representation is how clearly it portrays the nature of the variability associated with each variable. For example, we easily see from Fig 1.1 that the "center of action" for the Y_A data is somewhere around the group whose bar is centered around 75 (i.e., in the interval (74.51, 75.50)). Furthermore, most of the values of Y_A cluster in the 4 central groups centered around 74, 75, 76, and 77. In fact, 41 out of the 50 observations, or 82%, fall into these 4 groups; groups further away from the center of action (to the left as well as to the right) contribute less to the Y_A data. Similarly, Fig 1.2 shows that the "center of action" for the Y_B data is located somewhere around the group in the (71.51, 72.50) interval but it is not as sharply defined as it was with Y_A. Also the values of Y_B are more "spread out" and do not cluster as tightly around this central group.

The histogram also provides *quantitative* insight. For example, we see that 38 of the 50 Y_A observations (or 76%) are greater than 74.51; only 13 out of the 50 Y_B observations (or 26%) fall into this category. Also, exactly 0% of Y_A observations are less than or equal to 71.50 compared with 20 out of 50 observations (or a staggering 40%) of Y_B observations. Thus, if these data sets can be considered as representative of the overall performance of each process, then it is reasonable to conclude, for example, that there is a better chance of obtaining yields greater than 74.50 with process A than with process B (a 76% chance compared to a 26% chance). Similarly, while it is highly unlikely that process A will ever return yields less than 71.50, there is a not-insignificant chance (40%) that the yield obtained from process B will be less than 71.50. What is thus beginning to emerge are the faint outlines of

TABLE 1.5: Group classification and frequencies for the *inclusions* data

X	Frequency	Relative Frequency
0	22	0.367
1	23	0.383
2	11	0.183
3	1	0.017
4	2	0.033
5	1	0.017
6	0	0.000
TOTAL	60	1.000

a rigorous framework for characterizing and quantifying random variability, with the histogram providing this first glimpse.

It is important to note that the advantage provided by the histogram comes at the expense of losing the *individuality* of each observation. Having gone from 50 raw observations each to 8 groups for Y_A, and a slightly larger 12 groups for Y_B, there is clearly a loss of resolution: the individual identities of the original observations are no longer visible *from the histogram.* (For example, the identities of each of the 17 Y_A observations that make up the group in the interval (74.51,75.50) have been melded into that of a single, monolithic bar in the histogram.) But this is not necessarily a bad thing. As we demonstrate in upcoming chapters, a fundamental tenet of the probabilistic approach to dealing with randomly varying phenomena is an abandonment of the individual observation as the basis for theoretical characterization, in favor of an ensemble description. For now, it suffices to be able to see from this example that the clarity with which the histogram portrays data variability has been achieved by trading off the individual observation's identity for the ensemble identity of groups. But keep in mind that what the histogram offers is simply an alternative (albeit more informative) way of representing the same identical information contained in the data tables.

Let us now return to the second problem. In this case, the group classification and frequency distribution for the raw *inclusions* data is shown in Table 1.5. Let it not be lost on the reader that while the groups for the yield data sets were created from intervals of finite length, no such *quantization* is necessary for the *inclusions* data since in this case, the variable of interest, X, is naturally discrete. This fundamental difference between continuous variables (such as Y_A and Y_B) and discrete variables (such as X) will continue to surface at various stages in subsequent discussions.

The histogram for the *inclusions* data is shown in Fig 1.3 where several characteristics are now clear: for example, 75% of the glass sheets (45 out of 60) are either perfect or have only a single (almost inconsequential) inclusion; only

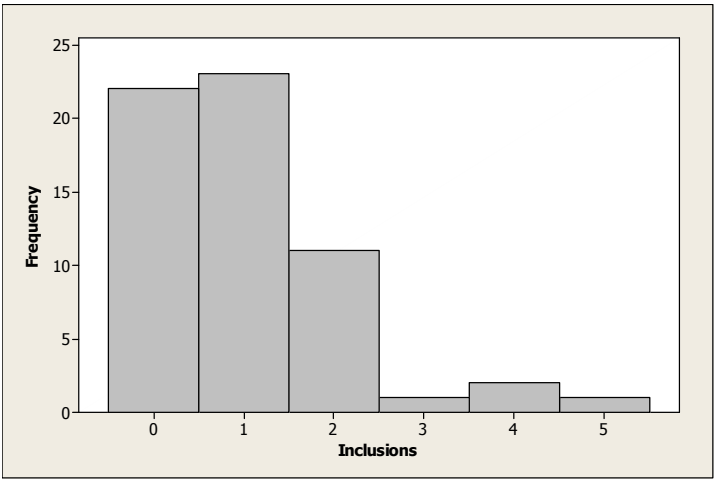


FIGURE 1.3: Histogram of inclusions data.

5% of the glass sheets (3 out of 60) have more than 3 inclusions, the remaining 95% have 3 or fewer; 93.3% (56 out of 60) have 2 or fewer inclusions. The important point is that such quantitative characteristics of the data variability (made possible by the histogram) is potentially useful for answering practical questions about what one can reasonably expect from this process.

1.3.2 Theoretical Distributions

How can the benefits of the histogram be consolidated into a useful tool for quantitative analysis of randomly varying phenomena? The answer: by appealing to a fundamental axiom of random phenomena: that conceptually, as more observations are made, the shape of the data histogram stabilizes, and tends to the form of the *theoretical distribution* that characterizes the random phenomenon in question, *in the limit as the total number of observations approaches infinity*. It is important to note that this concept does not necessarily require that an infinite number of observations actually be obtained in practice, even if this were possible. The essence of the concept is that an underlying theoretical distribution exists for which the frequency distribution represented by the histogram is but a "finite sample" approximation; that the underlying theoretical distribution is an ideal model of the particular phenomenon responsible for generating the finite number of observations contained in the current data set; and hence that this theoretical distribution provides a reasonable mathematical characterization of the random phenomenon.

As we show later, these theoretical distributions may be derived from first principles given sufficient knowledge regarding the underlying random phenomena. And, as the brief informal examination of the illustrative histograms

above indicates, these theoretical distributions can be used for various things. For example, even though we have not yet provided any concrete definition of the term *probability*, neither have we given any concrete justifications of its usage in this context, still from the discussion in the previous section, the reader can intuitively attest to the reasonableness of the following statements: "the probability that $Y_A \geq 74.5$ is ≈ 0.76"; or "the probability that $Y_B \geq 74.5$ is ≈ 0.26"; or "the probability that $X \leq 1$ is ≈ 0.75." Parts II and III are devoted to establishing these ideas more concretely and more precisely.

A Preview

It turns out that the theoretical distribution for each yield data set is:

$$f(y|\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(y-\mu)^2}{2\sigma^2}}; -\infty < y < \infty \tag{1.3}$$

which, when superimposed on each histogram, is shown in Fig 1.4 for Y_A, and Fig 1.5 for Y_B, when the indicated "characteristic parameters" are specified as $\mu = 75.52, \sigma = 1.43$ for Y_A, and $\mu = 72.47, \sigma = 2.76$ for Y_B.

Similarly, the theoretical distribution for the *inclusions* data is:

$$f(x|\lambda) = \frac{e^{-\lambda}\lambda^x}{x!}; x = 0, 1, 2, \ldots \tag{1.4}$$

where the characteristic parameter $\lambda = 1.02$ is the average number of *inclusions* in each glass sheet. In similar fashion to Eq (1.3), it also provides a theoretical characterization and quantification of the random phenomenon responsible for the variability observed in the *inclusions* data. From it we are able, for example, to compute the "theoretical probabilities" of observing $0, 1, 2, \ldots$, *inclusions* in any one glass sheet manufactured by this process. A plot of this theoretical probability distribution function is shown in Fig 1.6 (compare with the histogram in Fig 1.3).

The full detail of precisely what all this means is discussed in subsequent chapters; for now, this current brief preview serves the purpose of simply indicating how the expression in Eqs (1.3) and (1.4) provide a theoretical means of characterizing (and quantifying) the random phenomenon involved respectively in the yield data and in the inclusions data. Expressions such as this are called "probability distribution functions" (pdfs) and they provide the basis for rational analysis of random variability via the concept of "probability." Precisely what this concept of "probability" is, how it gives rise to pdfs, and how pdfs are used to solve practical problems and provide answers to the sorts of questions posed by these illustrative examples, constitute the primary focus of the remaining chapters in the book.

At this point, it is best to defer the rest of the discussion until when we revisit these two problems at appropriate places in upcoming chapters where we show that:

1. Y_A indeed may be considered as "greater" than Y_B, and in particular, that $Y_A - Y_B > 2$, up to a specific, quantifiable "degree of confidence."

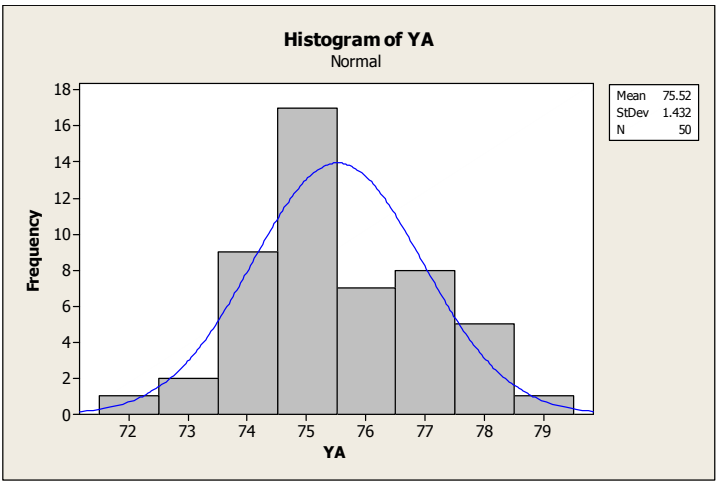


FIGURE 1.4: Histogram for Y_A data with superimposed theoretical distribution.

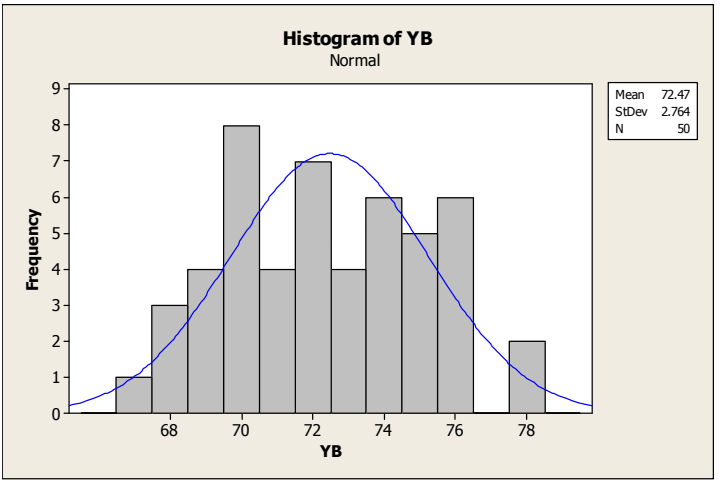


FIGURE 1.5: Histogram for Y_B data with superimposed theoretical distribution.

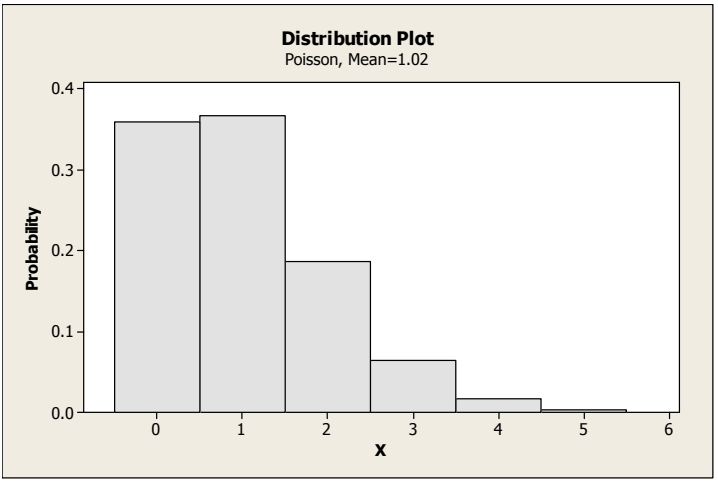


FIGURE 1.6: Theoretical probability distribution function for a Poisson random variable with parameter $\lambda = 1.02$. Compare with the inclusions data histogram in Fig 1.3.

2. There is in fact no evidence in the *inclusions* data to suggest that the process has deviated from its design target; i.e., that there is no reason to believe that $X \neq X^*$, again up to a specific, quantifiable "degree of confidence."

1.4 Summary and Conclusions

We have introduced two practical problems in this chapter to illustrate the complications caused by the presence of randomly varying phenomena in engineering problems. One problem involved determining which of two *continuous* variables is larger; the other involved determining if a *discrete* variable has deviated from its design target. Without the presence of random variability, each problem would ordinarily have been trivial to solve. However, with intrinsic variability that could not be idealized away, it became clear that special techniques capable of coping explicitly with randomly varying phenomena would be required to solve these problems satisfactorily. We did not solve the problems, of course (that is reserved for later); we simply provided an outline of a systematic approach to solving them, which required introducing some concepts that are to be explored fully later. As a result, the very brief introduction of the frequency distribution, the graphical histogram, and the theoretical distribution function was intended to serve merely as a preview of

upcoming detailed discussions concerning how randomly varying phenomena are analyzed systematically.

Here are some of the main points of the chapter again:

- The presence of random variability often complicates otherwise straightforward problems so that specialized solution techniques are required.
- Frequency distributions and histograms provide a particularly informative perspective of random variations intrinsic to experimental data.
- The probability distribution function—the theoretical limit to which the frequency distribution (and histogram) tends—provides the basis for systematic analysis of randomly varying phenomena.

REVIEW QUESTIONS

1. What decision is to be made in the yield improvement problem of Section 1.1?

2. What are the economic factors to be taken into consideration in deciding what to do with the yield improvement problem?

3. What is the essence of the yield improvement problem as discussed in Section 1.1?

4. What are some of the sources of variability associated with the process yields?

5. Why are the yield variables, Y_A and Y_B, *continuous* variables?

6. What single value is suggested as "intuitive" for representing a collection of n data points, $x_1, x_2, \ldots, x_n$?

7. What are some of the issues raised by entertaining the idea of representing the yield data sets with the arithmetic averages $\bar{y}_A$ and $\bar{y}_B$?

8. Why is the number of *inclusions* found on each glass sheet a discrete variable?

9. What are some sources of variability associated with the glass manufacturing process which may ultimately be responsible for the variability observed in the number of *inclusions*?

10. What is a frequency distribution and how is it obtained from raw data?

11. Why will bin size affect the appearance of a frequency distribution?

12. What is a histogram and how is it obtained from data?

13. What is the primary advantage of a histogram over a table of raw data?

14. What is the relationship between a histogram and a theoretical distribution?

15. What are the expressions in Eqs (1.3) and (1.4) called? These equations provide the basis for what?

EXERCISES

Section 1.1

1.1 The variance of a collection of n data points, $y_1, y_2, \ldots, y_n$, is defined as:

$$s^2 = \frac{\sum_{i=1}^{n}(y_i - \bar{y})^2}{n-1} \tag{1.5}$$

where $\bar{y}$ is the arithmetic average of the data set. From the yield data in Table 1.1, obtain the variances s_A^2 and s_B^2 for the Y_A and Y_B data sets, respectively. Which is greater, s_A^2 or s_B^2?

1.2 Even though the data sets in Table 1.1 were not generated in pairs, obtain the 50 differences,

$$d_i = y_{A_i} - y_{B_i};\ i = 1, 2, \ldots, 50 \tag{1.6}$$

for corresponding values of Y_A and Y_B as presented in this table. Obtain a histogram of d_i and compute the arithmetic average,

$$\bar{d} = \frac{1}{n}\sum_{i=1}^{n} d_i \tag{1.7}$$

What do these results suggest about the possibility that Y_A may be greater than Y_B?

1.3 A set of theoretical results to be established later (see Chapter 4 Exercises) state that, for d_i and $\bar{d}$ defined in Eq (1.7), and variance s^2 defined in Exercise 1,

$$\bar{d} = \bar{y}_A - \bar{y}_B \tag{1.8}$$

$$s_d^2 = s_A^2 + s_B^2 \tag{1.9}$$

Confirm these results specifically for the data in Table 1.1.

Section 1.2

1.4 From the data in Table 1.2, obtain s_x^2, the variance of the inclusions.

1.5 The random variable, X, representing the number of inclusions, is purported to be a Poisson random variable (see Chapter 8). If true, then the average, $\bar{x}$, and

variance, s_x^2, are theoretically equal. Compare the values computed for these two quantities from the data set in Table 1.2. What do these results suggest about the possibility that X may in fact be a Poisson random variable?

Section 1.3

1.6 Using a bin size of 0.75, obtain relative frequencies for Y_A and Y_B data and the corresponding histograms. Repeat this exercise for a bin size of 2.0. Compare these two sets of histograms with the corresponding histograms in Figs 1.1 and 1.2.

1.7 From the frequency distribution in Table 1.3 and the values computed for the average, $\bar{y}_A$, and variance, s_A^2 of the yield data set, Y_A, determine the percentage of the data contained in the interval $\bar{y}_A \pm 1.96 s_A$, where s_A is the positive square root of the variance, s_A^2.

1.8 Repeat Exercise 1.7 for the Y_B data in Table 1.4. Determine the percentage of the data contained in the interval $\bar{y}_B \pm 1.96 s_B$.

1.9 From Table 1.5 determine the value of x such that only 5% of the data exceeds this value.

1.10 Using $\mu = 75.52$ and $\sigma = 1.43$, compute theoretical values of the function in Eq (1.3) at the center points of the frequency groups for the Y_A data in Table 1.3; i.e., for $y = 72, 73, \ldots, 79$. Compare these theoretical values with the corresponding relative frequency values.

1.11 Repeat Exercise 1.10 for Y_B data and Table 1.4 this time with $\mu = 72.47$ and $\sigma = 2.76$ and for $y = 67, 68, 69, \ldots, 79$

1.12 Using $\lambda = 1.02$, compute theoretical values of the function $f(x|\lambda)$ in Eq (1.4) at $x = 0, 1, 2, \ldots 6$ and compare with the corresponding relative frequency values in Table 1.5.

APPLICATION PROBLEMS

1.13 The data set in the table below is the time (in months) from receipt to publication (sometimes known as *time-to-publication*) of 85 papers published in the January 2004 issue of a leading chemical engineering research journal.

(i) Generate a histogram of this data set. Comment on the "shape" of this histogram and why, from the nature of the variable in question, such a shape may not be surprising.

(ii) From the histogram of the data, what is the "most popular" time-to-publication, and what fraction of the papers took longer than this to publish?

19.2	15.1	9.6	4.2	5.4
9.0	5.3	12.9	4.2	15.2
17.2	12.0	17.3	7.8	8.0
8.2	3.0	6.0	9.5	11.7
4.5	18.5	24.3	3.9	17.2
13.5	5.8	21.3	8.7	4.0
20.7	6.8	19.3	5.9	3.8
7.9	14.5	2.5	5.3	7.4
19.5	3.3	9.1	1.8	5.3
8.8	11.1	8.1	10.1	10.6
18.7	16.4	9.8	10.0	15.2
7.4	7.3	15.4	18.7	11.5
9.7	7.4	15.7	5.6	5.9
13.7	7.3	8.2	3.3	20.1
8.1	5.2	8.8	7.3	12.2
8.4	10.2	7.2	11.3	12.0
10.8	3.1	12.8	2.9	8.8

1.14 Refer to Problem 1.13. Let each raw data entry in the data table be x_i.
(i) Generate a set of 85 "sample average publication time," y_i, from 20 consecutive times as follows:

$$y_1 = \frac{1}{20}\sum_{i=1}^{20} x_i \tag{1.10}$$

$$y_2 = \frac{1}{20}\sum_{i=2}^{21} x_i \tag{1.11}$$

$$y_3 = \frac{1}{20}\sum_{i=3}^{22} x_i \tag{1.12}$$

$$\ldots = \ldots$$

$$y_j = \frac{1}{20}\sum_{i=j}^{20+(j-1)} x_i \tag{1.13}$$

For values of $j \geq 66$, y_j should be obtained by replacing $x_{86}, x_{87}, x_{88}, \ldots$, which do not exist, with $x_1, x_2, x_3, \ldots$, respectively (i.e., for these purposes treat the given x_i data like a "circular array"). Plot the histogram for this generated y_i data and compare the shape of this histogram with that of the original x_i data.
(ii) Repeat part (i) above, this time for z_i data generated from:

$$z_j = \frac{1}{20}\sum_{i=j}^{20+(j-1)} y_i \tag{1.14}$$

for $j = 1, 2, \ldots, 85$. Compare the histogram of the z_i data with that of the y_i data and comment on the effect of "averaging" on the shape of the data histograms.

1.15 The data shown in the table below is a four-year record of the number of "recordable" safety incidents occurring at a plant site each month.

1	0	0	0	2	2	0	0	0	1	0	1
0	1	0	1	0	0	0	0	0	0	0	1
2	2	0	1	2	0	1	2	1	1	0	0
0	1	0	0	0	0	0	0	1	0	0	0

(i) Find the average number of safety incidents per month and the associated variance. Construct a frequency table of the data and plot a histogram.

(ii) From the frequency table and the histogram, what can you say about the "chances" of obtaining each of the following observations, where x represents the number of observed safety incidents per month: $x = 0, x = 1, x = 2, x = 3, x = 4$ and $x = 5$?

(iii) Consider the postulate that a reasonable model for this phenomenon is:

$$f(x) = \frac{e^{-0.5}0.5^x}{x!} \tag{1.15}$$

where $f(x)$ represents the theoretical "probability" of recording exactly x safety incidents per month. How well does this model fit the data?

(iv) Assuming that this is a reasonable model, discuss how you would use it to answer the question: *If, over the most recent four-month period, the plant recorded 1, 3, 2, 3 safety incidents respectively, is there evidence that there has been a "real increase" in the number of safety incidents*?

1.16 The table below shows a record of the "before" and "after" weights (in pounds) of 20 patients enrolled in a clinically supervised 10-week weight-loss program.

Patient #	1	2	3	4	5	6	7	8	9	10
Before Wt (lbs)	272	319	253	325	236	233	300	260	268	276
After Wt (lbs)	263	313	251	312	227	227	290	251	262	263
Patient #	11	12	13	14	15	16	17	18	19	20
Before Wt (lbs)	215	245	248	364	301	203	197	217	210	223
After Wt (lbs)	206	235	237	350	288	195	193	216	202	214

Let X_B represent the "before" weight and X_A the "after" weight.

(i) Using the same bin size for each data set, obtain histograms for the X_B and X_A data and plot both on the same graph. Strictly on the basis of a visual inspection of these histograms, what can you say about the effectiveness of the weight-loss program in achieving its objective of assisting patients to lose weight?

(ii) Define the difference variable, $D = X_B - X_A$, and from the given data, obtain and plot a histogram for this variable. Again, strictly from a visual inspection of this histogram, what can you say about the effectiveness of the weight-loss program?

1.17 The data shown in the following table is from an "Assisted Reproductive Technologies" clinic where a cohort of 100 patients under the age of 35 years (the "younger" group), and another cohort, 35 years and older (the "older" group), each received 5 embryos in an in-vitro fertilization (IVF) treatment cycle.

x No. of live births in a delivered pregnancy	y_O Total no. of "older patients" (out of 100) with pregnancy outcome x	y_Y Total no. of "younger patients" (out of 100) with pregnancy outcome x
0	32	8
1	41	25
2	21	35
3	5	23
4	1	8
5	0	1

The data shows x, the number of live births per delivered pregnancy, along with how many in each group had the pregnancy outcome of x. For example, the first entry indicates that the IVF treatment was unsuccessful for 32 of the "older" patients, with the corresponding number being 8 for the "younger" patients; 41 "older" patients delivered singletons, compared with 25 for the younger patients; 21 older patients and 35 younger patients each delivered twins; etc. Obtain a *relative* frequency distribution for these data sets and plot the corresponding histograms. Determine the average number of live births per delivered pregnancy for each group and compare these values. Comment on whether or not these data sets indicate that the outcomes of the IVF treatments are different for these two groups.

Chapter 2

Random Phenomena, Variability, and Uncertainty

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Through the great beneficence of Providence,
what is given to be foreseen in the general sphere of masses
escapes us in the confined sphere of individuals.

Joannè-Erhard Valentin-Smith (1796–1891)

When John Stuart Mills stated in his 1862 book, *A System of Logic: Ratiocinative and Inductive*, that

> *the very events which in their own nature appear most capricious and uncertain, and which in any individual case no attainable degree of knowledge would enable us to foresee, occur, when considerable numbers are taken into account, with a degree of regularity approaching to mathematical,*

he was merely articulating—astutely for the time—the then-radical, but now well-accepted, concept that randomness in scientific observation is not a synonym for disorder; it is *order of a different kind.* The more familiar kind

of order informs "determinism": the concept that, with sufficient mechanistic knowledge, all physical phenomena are entirely predictable and thus describable by precise mathematical equations. But even classical physics, that archetypal deterministic science, had to make room for this *other* kind of order when quantum physicists of the 1920s discovered that fundamental particles of nature exhibit irreducible uncertainty (or "chance") in their locations, movements, and interactions. Today, most contemporary scientists and engineers are, by training, conditioned to accept both determinism and randomness as intrinsic aspects of the experiential world. The problem, however, is that to many, the basic characteristics of random phenomena and their "order of a different kind" still remain somewhat unfamiliar at a fundamental level.

This chapter is devoted to an expository examination of randomly varying phenomena. Its primary purpose is to introduce the reader to the central characteristic of order-in-the-midst-of-variability, and the sort of analysis this trait permits, *before* diving headlong into a formal study of probability and statistics. The premise of this chapter is that a true appreciation of the nature of randomly varying phenomena at a fundamental level is indispensable to the sort of clear understanding of probability and statistics that will protect the diligent reader from all-too-common misapplication pitfalls.

2.1 Two Extreme Idealizations of Natural Phenomena

2.1.1 Introduction

In classical physics, the distance, x (in meters), traveled in t seconds by an object launched with an initial velocity, u m/s, and which accelerates at a m/s^2, is known to be given by the expression:

$$x = ut + \frac{1}{2}at^2 \tag{2.1}$$

This is a "deterministic" expression: it consistently and repeatably produces the same result every time identical values of the variables u, a, and t are specified. The same is true for the expression used in engineering to determine Q, the rate of heat loss from a house, say in the middle of winter, when the total exposed surface area is A m^2, the inside temperature is T_i K, the outside temperature is T_o K, and the combined heat transfer characteristics of the house walls, insulation, etc., is represented by the so-called "overall heat transfer coefficient" U, W/m^2K, i.e.,

$$Q = UA(T_i - T_o) \tag{2.2}$$

The rate of heat loss is determined precisely and consistently for any given specific values of each entity on the right hand side of this equation.

The concept of determinism, that the phenomenon in question is *precisely* determinable in every relevant detail, is central to much of science and engineering and has proven quite useful in analyzing real systems, and in solving practical problems—whether it is computing the trajectory of rockets for launching satellites into orbit, installing appropriate insulation for homes, or designing chemical reactors. However, any assumption of strict determinism in nature is implicitly understood as a convenient idealization resulting from neglecting certain details considered non-essential to the core problem. For example, the capriciousness of the wind and its various and sundry effects have been ignored in Eqs (2.1) and (2.2): no significant wind resistance (or assistance) in the former, negligible convective heat transfer in the latter.

At the other extreme is "randomness," where the relevant details of the phenomenon in question are indeterminable precisely; repeated observations under identical conditions produce different and randomly varying results; and the observed random variability is essential to the problem and therefore cannot be idealized away. Such is the case with the illustrative problems in Chapter 1 where in one case, the yield obtained from each process may be idealized as follows:

$$y_{Ai} = \eta_A + \epsilon_{Ai} \tag{2.3}$$

$$y_{Bi} = \eta_B + \epsilon_{Bi} \tag{2.4}$$

with η_A and η_B representing the "true" but unknown yields obtainable from processes A and B respectively, and ϵ_{Ai} and ϵ_{Bi} representing the superimposed randomly varying component—the sources of the random variability evident in each observation y_{Ai} and y_{Bi}. Identical values of η_A do not produce identical values of y_{Ai} in Eq (2.3); neither will identical values of η_B produce identical values of y_{Bi} in Eq (2.4). In the second case of the glass process and the number of *inclusions* per square meter, the idealization is:

$$x_i = \xi + \epsilon_i \tag{2.5}$$

where ξ is the "true" number of inclusions associated with the process and ϵ_i is the superimposed random component responsible for the observed randomness in the actual number of inclusion x_i found on each individual glass sheet upon inspection.

These two perspectives, "determinism" and "randomness," are thus two opposite idealizations of natural phenomena, the former when deterministic aspects of the phenomenon are considered to be overwhelmingly dominant over any random components, the latter case when the random components are dominant and central to the problem. The principles behind each conceptual idealization, and the analysis technique appropriate to each, are now elucidated with a chemical engineering illustration.

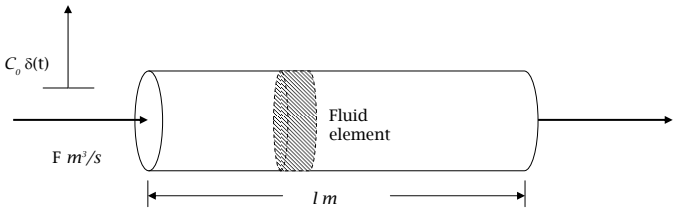


FIGURE 2.1: Schematic diagram of a plug flow reactor (PFR).

2.1.2 A Chemical Engineering Illustration

Residence time, the amount of time a fluid element spends in a chemical reactor, is an important parameter in the design of chemical reactors. We wish to consider residence times in two classic reactor configurations: the plug flow reactor (PFR) and the continuous stirred tank reactor (CSTR).

Determinism and the PFR

The plug flow reactor (PFR) is a hollow tube in which reactants that are introduced at one end react as the fluid elements traverse the length of the tube and emerge at the other end. The name comes from the idealization that fluid elements move through as "plugs" with no longitudinal mixing (see Fig 2.1).

The PFR assumptions (idealizations) may be stated as follows:

- The reactor tube (l m long) has a uniform cross-sectional area, A m^2.
- Fluid elements move in "plug flow" with a constant velocity, v m/s, so that the velocity profile is flat.
- The flow rate through the reactor is constant at F m^3/s.

Now consider that at time $t = 0$ we instantaneously inject a bolus of red dye of concentration C_0 moles/m^3 into the inlet stream. The following question is of interest in the study of residence time distributions in chemical reactor design:

> How much time does each molecule of red dye spend in the reactor, if we could label them all and observe each one at the reactor exit?

Because of the plug flow idealization, each fluid element moves through the reactor with a constant velocity given by:

$$v = \left(\frac{F}{A}\right) m/s \tag{2.6}$$

and it will take precisely

$$\theta = \frac{l}{v} = \left(\frac{lA}{F}\right) secs \tag{2.7}$$

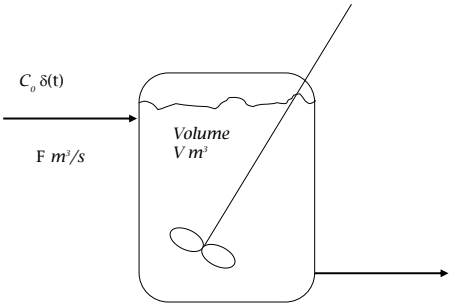


FIGURE 2.2: Schematic diagram of a continuous stirred tank reactor (CSTR).

for each dye element to traverse the reactor. Hence, θ, the residence time for an ideal plug flow reactor (PFR) is a deterministic quantity because its value is exactly and precisely determinable from Eq (2.7) given F, A and l.

Keep in mind that the determinism that informs this analysis of the PFR residence time arises directly as a consequence of the central plug flow idealization. Any departures from such idealization, especially the presence of significant axial dispersion (leading to a non-flat fluid velocity profile), will result in dye molecules no longer arriving at the outlet at precisely the same time.

Randomness and the CSTR

With the continuous stirred tank reactor (CSTR), the reactant stream continuously flows into a tank that is vigorously stirred to ensure uniform mixing of its content, while the product is continuously withdrawn from the outlet (see Fig 2.2). The assumptions (idealizations) in this case are:

- The reactor tank has a fixed, constant volume, $V\ m^3$.

- The contents of the tank are perfectly mixed.

Once again, let us consider that a bolus of red dye of concentration C_0 moles/m^3 is instantaneously injected into the inlet stream at time $t = 0$; and again, ask: how much time does each molecule of red dye spend in the reactor? Unlike with the plug flow reactor, observe that it is impossible to answer this question *à-priori*, or precisely: because of the vigorous stirring of the reactor content, some dye molecules will exit almost instantaneously; others will stay longer, some for a *very* long time. In fact, it can be shown that theoretically, $0 < \theta < \infty$. Hence in this case, θ, the residence time, is a randomly varying quantity that can take on a range of values from 0 to ∞; it cannot therefore be adequately characterized as a single number. Notwithstanding, as all chemical

engineers know, the random phenomenon of residence times for ideal CSTR's can, and has been, analyzed systematically (see for example, Hill, 1977[1]).

Theoretical Analysis of the Ideal CSTR's Residence Time

Even though based on chemical engineering principles, the results of the analysis we are about to discuss have fundamental implications for the general nature of the order present in the midst of random variability encountered in other applications, and how such order provides the basis for analysis. (As an added bonus, this analysis also provides a non-probabilistic view of ideas usually considered the exclusive domain of probability.)

By carrying out a material balance around the CSTR, (i.e., that the rate of accumulation of mass within a prescribed volume must equal the difference between the rate of input and the rate of output) it is possible to develop a mathematical model for this process as follows: If the volumetric flow rate into and out of the reactor are equal and given by $F\ m^3/s$, if $C(t)$ represents the molar concentration of the dye in the well-mixed reactor, then by the assumption of perfect mixing, this will also be the dye concentration at the exit of the reactor. The material balance equation is:

$$V\frac{dC}{dt} = FC_{in} - FC \tag{2.8}$$

where C_{in} is the dye concentration in the inlet stream. If we define the parameter τ as

$$\tau = \frac{V}{F} \tag{2.9}$$

and note that the introduction of a bolus of dye of concentration C_0 at $t = 0$ implies:

$$C_{in} = C_0\delta(t) \tag{2.10}$$

where $\delta(t)$ is the Dirac delta function, then Eq (2.8) becomes:

$$\tau\frac{dC}{dt} = -C + C_0\delta(t) \tag{2.11}$$

a simple, linear first order ODE whose solution is:

$$C(t) = \frac{C_0}{\tau}e^{-t/\tau} \tag{2.12}$$

If we now define as $f(\theta)$, the *instantaneous* fraction of the initial number of injected dye molecules exiting the reactor at time $t = \theta$ (those with residence time θ), i.e.,

$$f(\theta) = \frac{C(\theta)}{C_0} \tag{2.13}$$

[1]C.G. Hill, Jr. (1977). *An Introduction to Chemical Engineering Kinetics and Reactor Design*, Wiley, New York, pp 388–396.

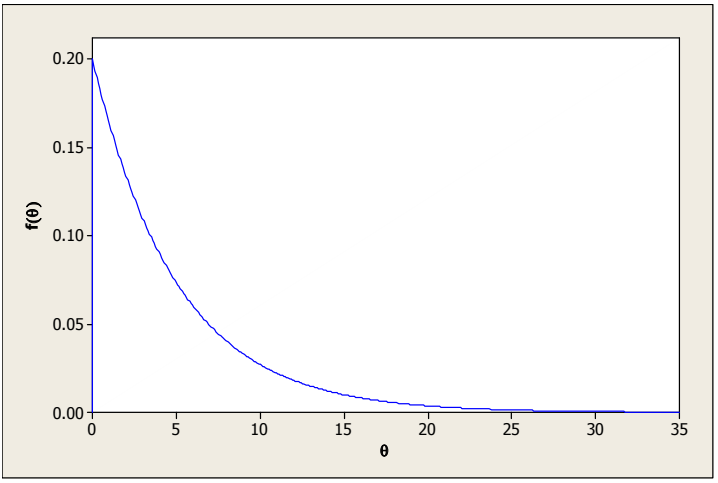


FIGURE 2.3: Instantaneous residence time distribution function for the CSTR with $\tau = 5$.

we obtain immediately from Eq (2.12) that

$$f(\theta) = \frac{1}{\tau} e^{-\theta/\tau} \tag{2.14}$$

recognizable to all chemical engineers as the familiar "exponential" *instantaneous* residence time distribution function for the ideal CSTR. The reader should take good note of this expression: it shows up a few more times and in various guises in subsequent chapters. For now, let us observe that, even though (a) the residence time for a CSTR, θ, exhibits random variability, potentially able to take on values between 0 and ∞ (and is therefore not describable by a single value); so that (b) it is therefore impossible to determine with absolute certainty precisely when any individual dye molecule will leave the reactor; even so (c) the function, $f(\theta)$, shown in Eq (2.14), mathematically characterizes the behavior of the entire ensemble of dye molecules, but in a way that requires some explanation:

1. It represents how the residence times of fluid particles in the well-mixed CSTR are "distributed" over the range of possible values $0 < \theta < \infty$ (see Fig 2.3).

2. This distribution of residence times is a well-defined, well-characterized function, but it is not a description of the precise amount of time a particular individual dye molecule will spend in the reactor; rather it is a description of how many (or what fraction) of the *entire collection* of dye molecules will spend what amount of time in the reactor. For example, in broad terms, it indicates that a good fraction of the molecules have relatively short residence times, exiting the reactor quickly; a much

smaller but non-zero fraction have relatively long residence times. It can also provide more precise statements as follows.

3. From this expression (Eq (2.14)), we can determine the fraction of dye molecules that have remained in the reactor for an amount of time less than or equal to some time t (i.e., molecules exiting the reactor with "age" less than or equal to t); we do this by integrating $f(\theta)$ with respect to θ, as follows, to obtain

$$F(t) = \int_0^t \frac{1}{\tau} e^{-\theta/\tau} d\theta = \left(1 - e^{-t/\tau}\right) \tag{2.15}$$

from which we see that $F(0)$, the fraction of dye molecules with age less than or equal to zero is exactly zero: indicating the intuitively obvious that, no matter how vigorous the mixing, each dye molecule spends at least a finite, non-zero, amount of time in the reactor (no molecule exits instantaneously upon entry).

On the other hand, $F(\infty) = 1$, since

$$F(\infty) = \int_0^\infty \frac{1}{\tau} e^{-\theta/\tau} d\theta = 1 \tag{2.16}$$

again indicating the obvious: if we wait long enough, *all* dye molecules will eventually exit the reactor as $t \to \infty$. In other words, the fraction of molecules exiting the reactor with age less than ∞ is exactly 1.

4. Since the fraction of molecules that will have remained in the reactor for an amount of time less than or equal to t is $F(t)$, and the fraction that will have remained in the reactor for less than or equal to $t + \Delta t$ is $F(t + \Delta t)$, then the fraction with residence time in the infinitesimal interval between t and $(t + \Delta t)$ is given by:

$$\Phi[t \leq \theta \leq (t + \Delta t)] = F(t + \Delta t) - F(t) = \int_t^{t+\Delta t} \frac{1}{\tau} e^{-\theta/\tau} d\theta \tag{2.17}$$

which, for very small Δt, simplifies to:

$$\Phi[t \leq \theta \leq (t + \Delta t)] \approx f(t)\Delta t \tag{2.18}$$

5. And finally, the "average residence time" may be determined from the expression in Eq (2.14) (and Eq (2.16)) as:

$$\bar{\theta} = \frac{\int_0^\infty \frac{1}{\tau}\theta e^{-\theta/\tau} d\theta}{\int_0^\infty \frac{1}{\tau} e^{-\theta/\tau} d\theta} = \frac{\frac{1}{\tau}\int_0^\infty \theta e^{-\theta/\tau} d\theta}{1} = \tau \tag{2.19}$$

where the numerator integral is evaluated via integration by parts. Observe from the definition of τ above (in Eq (2.9)) that this result makes perfect sense, strictly from the physics of the problem: particles in a stream flowing at the rate F m^3/s through a well-mixed reactor of volume V m^3, will spend an average of $V/F = \tau$ seconds in the reactor.

We now observe in conclusion two important points. (i) Even though at no point in the preceding discussion have we made any overt or explicit appeal to the concepts of probability, the unmistakable fingerprints of probability are evident all over (as upcoming chapters demonstrate concretely, but perhaps already recognizable to those with some familiarity with such concepts). (ii) Nevertheless, this characterizing model in Eq (2.14) was made possible via first-principles knowledge of the underlying phenomenon. This is a central characteristic of random phenomena: that appropriate theoretical characterizations are almost always possible in terms of ideal ensemble models of the observed variability dictated by the underlying phenomenological mechanism.

2.2 Random Mass Phenomena

2.2.1 Defining Characteristics

In such diverse areas as actuarial science, biology, chemical reactors, demography, economics, finance, genetics, human mortality, manufacturing quality assurance, polymer chemistry, etc., one repeatedly encounters a surprisingly common theme whereby phenomena which, on an individual level, appear entirely unpredictable, are well-characterized as ensembles (as demonstrated above with residence time distribution in CSTRs). For example, as far back as 1662, in a study widely considered to be the genesis of population demographics and of modern actuarial science by which insurance premiums are determined today, the British haberdasher, John Graunt (1620–1674), had observed that the number of deaths and the age at death in London were surprisingly predictable for the entire population even though it was impossible to predict which individual would die when and in what manner. Similarly, while the number of monomer molecules linked together in any polymer molecule chain varies considerably, how many chains of a certain length a batch of polymer product contains can be characterized fairly predictably.

Such natural phenomena noted above have come to be known as *random mass phenomena*, with the following defining characteristics:

1. Individual observations appear irregular because it is not possible to predict each one with certainty.

2. The ensemble or aggregate of all possible outcomes is regular, well-characterized, and determinable.

3. The underlying phenomenological mechanisms accounting for the "nature" and occurrence of the specific observations determines the character of the ensemble.

4. Such phenomenological mechanisms may be known mechanistically (as

was the case with the CSTR), or its manifestation may only be determined from data (as was the case with John Graunt's mortality tables of 1662).

This fortunate circumstance—aggregate predictability amidst individual irregularities—is why the primary issue with random phenomena analysis boils down to how to use ensemble descriptions and characterization to carry out systematic analysis of the behavior of individual observations.

2.2.2 Variability and Uncertainty

While ensemble characterizations provide a means of dealing systematically with random mass phenomena, many practical problems still involve making decisions about *specific*, inherently unpredictable, outcomes. For example, the insurance company still has to decide what premium to charge each individual on a person-by-person basis. When decisions must be made about specific outcomes of random mass phenomena, uncertainty is an inevitable consequence of the inherent variability. Furthermore, the extent or degree of variability directly affects the degree of uncertainty: tighter clustering of possible outcomes implies less uncertainty, whereas a broader distribution of possible outcomes implies more uncertainty. The most useful mathematical characterization of ensembles must therefore permit not only systematic analysis, but also a rational quantification of the degree of variability inherent in the ensemble, and the resulting uncertainty associated with each individual observation as a result.

2.2.3 Practical Problems of Interest

Let x_i represent individual observations, $i = 1, 2, \ldots, n$, from a random mass phenomenon; let X be the actual variable of interest, different and distinct from x_i, this latter being merely one out of many other possible realizations of X. For example, X can be the number of live births delivered by a patient after a round of in-vitro fertilization treatment, a randomly varying quantity; whereas $x_i = 2$ (i.e., twins) is the specific outcome observed for a specific patient after a specific round of treatment. For now, let the aggregate description we seek be represented as $f(x)$ (see for example, Eq (2.14) for the CSTR residence time); what this is and how it is obtained is discussed later. In practice, only data in the form of $\{x_i\}_{i=1}^n$ observations is available. The desired aggregate description, $f(x)$, must be understood in its proper context as a descriptor of the (possibly infinite) collection of all possible outcomes of which the observed data is only a "sample." The fundamental problems of random phenomena analysis may now be stated formally as follows:

1. Given $\{x_i\}_{i=1}^n$ what can we say about the complete $f(x)$?
2. Given $f(x)$ what can we say about the specific x_i values (both the observed $\{x_i\}_{i=1}^n$ and the yet unobserved)?

Embedded in these questions are the following affiliated questions that arise as a consequence. (a) How was $\{x_i\}_{i=1}^n$ obtained in (1); will the procedure for obtaining the data affect how well we can answer question 1? (b) How was $f(x)$ determined in (2)?

Subsequent chapters are devoted to dealing with these fundamental problems systematically and in greater detail.

2.3 Introducing Probability

2.3.1 Basic Concepts

Consider the prototypical random phenomenon for which the individual observation (or outcome) is not known with certainty *à-priori*, but the complete totality of *all* possible observations (or outcomes) has been (or can be) compiled. Now consider a framework that *assigns* to each individual member of this collection of possible outcomes, a real-valued number between 0 and 1 that represents *the probability of its occurrence*, such that:

1. An outcome that is certain to occur is assigned the number 1.
2. An outcome that is certain *not* to occur is assigned the number 0.
3. Any other outcome falling between these two extremes is assigned a number that reflects the extent or degree of certainty (or uncertainty) associated with its occurrence.

Notice how this represents a shift in focus from the individual outcome itself to *the probability of its occurrence*. Using precise definitions and terminology, along with tools of set theory, set functions and real analysis, we show in the chapters in Part II how to develop the machinery for the theory of probability, and the emergence of a compact functional form indicating how the probabilities of occurrence are "distributed" over all possible outcomes. The resulting "probability distribution function" becomes the primary vehicle for analyzing the behavior of random phenomena.

For example, the phenomenon of "inclusions" in manufactured glass sheets discussed in Chapter 1 is well-characterized by the following probability distribution function (pdf) which indicates the probability of observing exactly x "inclusions" on a glass sheet as

$$f(x) = \frac{e^{-\lambda}\lambda^x}{x!}; x = 0, 1, 2, \ldots \tag{2.20}$$

a pdf with a single "parameter," λ, characteristic of the manufacturing process used to produce the glass sheets. (As shown later, λ is the mean number of

TABLE 2.1: Computed probabilities of occurrence of various number of *inclusions* for $\lambda = 2$ in Eq (2.20)

x = No. of *Inclusions*	$f(x)$ Prob. of Occurrence
0	0.135
1	0.271
2	0.271
3	0.180
4	0.090
5	0.036
⋮	⋮
8	0.001
≥ 9	0.000

"inclusions" on a glass sheet.) Even though we do not know precisely how many "inclusions" will be found on the next glass sheet inspected in the QC lab, given the parameter λ, we can use Eq (2.20) to make statements about the probabilities of individual occurrences. For instance, if $\lambda = 2$ for a certain process, Eq (2.20) allows us to state that the probability of finding a perfect glass sheet with no "inclusions" in the products made by this process (i.e., $x = 0$) is 0.135; or that the probability of finding 1 "inclusion" is 0.227, coincidentally the same as the probability of finding 2 "inclusions"; or that there is a vanishingly small probability of finding 9 or more "inclusions" in this production facility. The complete set of probabilities computed from Eq (2.20) is shown in Table 2.1.

2.3.2 Interpreting Probability

There always seems to be a certain amount of debate over the meaning, definition, and interpretation of probability. This is perhaps due to a natural predisposition toward confusing a *conceptual entity* with *how* a numerical value is determined for it. For example, from a certain perspective, "temperature," as a conceptual entity in Thermodynamics, is a real number *assigned* to an object to indicate its degree of "hotness"; it is distinct from *how* its value is determined (by a thermometer, thermocouple, or any other means). The same is true of "mass," a quantity assigned in Mechanics to a body to indicate how much matter it contains and how heavy it will be in a gravitational field; or "distance" assigned in geometry to indicate the closeness of two points in a geometric space. The practical problem of how to determine numerical values for these quantities, even though important in its own right, is a separate issue entirely.

This is how probability should be understood: it is simply a quantity that

is *assigned* to indicate the degree of uncertainty associated with the occurrence of a particular outcome. How a numerical value is determined for the probability of the occurrence of a particular outcome is a separate issue that will depend on the specific problem at hand. To illustrate, consider temperature *the conceptual quantity*; while a thermometer capable of measuring temperature to within half a degree will suffice in one specific case, a more precise device, such as a thermocouple, may be required in another case, and an optical pyrometer for yet another case. Whatever the case, under no circumstance should the *device* employed to determine the numerical value usurp the role of, or become the surrogate for, "temperature" the quantity. This is important in properly interpreting probability, the conceptual entity: how an "appropriate" value is to be determined for probability, an important practical problem in its own right, should not be confused with the quantity itself.

With these ideas in mind, let us now consider several standard perspectives of probability that have evolved over the years. These are best understood as various techniques for *how* numerical values are determined rather than *what* probability is.

Classical (À-Priori) Probability

Consider a random phenomenon for which the total number of possible outcomes is known to be N, all of which are equally likely; of these, let N_A be the number of outcomes in which A is observed (i.e., outcomes that are "favorable" to A). Then according to the classical (or *à-priori*) perspective, the probability of the occurrence of outcome A is defined as

$$P(A) = \frac{N_A}{N} \tag{2.21}$$

For example, in tossing a single perfect die once, the probability of observing a 3 is, according to this viewpoint, evaluated as 1/6, since the total number of possible outcomes is 6 of which only 1 is favorable to the desired observation of 3. Similarly, if B is the outcome that one observes an odd number of dots, then $P(B) = 3/6 = 0.5$.

Observe that according to this view, no experiments have been performed yet; the formulation is based entirely on an *à-priori* enumeration of N and N_A. However, this intuitively appealing perspective is not always applicable:

- What if all the outcomes are not equally likely?
- How about random phenomena whose outcomes cannot be characterized as cleanly in this fashion, say, for example, the prospect of a newly purchased refrigerator lasting for 25 years without repair? Or the prospect of snow falling on a specific April day in Wisconsin?

What Eq (2.21) provides is an intuitively appealing (and theoretically sound) means of *determining an appropriate value* for $P(A)$; but it is restricted only to those circumstances where the random phenomenon in question is characterized in such a way that N and N_A are natural and easy to identify.

Relative Frequency (À-Posteriori) Probability

On the opposite end of the spectrum from the *à-priori* perspective is the following alternative: consider an "experiment" that is repeated n times under identical conditions, where the outcomes involving A have been observed to occur n_A times. Then, *à-posteriori*, the probability of the occurrence of outcome A is defined as

$$P(A) = \lim_{n\to\infty} \frac{n_A}{n} \tag{2.22}$$

The appeal of this viewpoint is not so much that it is just as intuitive as the previous one, but that it is also empirical, making no assumptions about equal likelihood of outcomes. It is based on the actual performance of "experiments" and the actual *à-posteriori* observation of the relative frequency of occurrences of the desired outcome. This perspective provides a prevalent interpretation of probability as the "theoretical" value of long range relative frequencies. In fact, this is what motivates the notion of the theoretical distribution as the limiting form to which the empirical frequency distribution tends with the acquisition of increasing amounts of data.

However, this perspective also suffers from some limitations:

- How many trials, n, is sufficient for Eq (2.22) to be useful in practice?
- How about random phenomena for which the desired outcome does not lend itself to repetitive experimentation under identical conditions, say, for example, the prospect of snow falling on a specific April day in Wisconsin? Or the prospect of your favorite team winning the basketball championship next year?

Once again, these limitations arise primary because Eq (2.22) is simply just another means of *determining an appropriate value* for $P(A)$ that happens to be valid only when the random phenomenon is such that the indicated repeated experimentation is not only possible and convenient, but for which, in practice, truncating after a "sufficiently large" number of trials to produce a finite approximation presents no conceptual dilemma. For example, after tossing a coin 500 times and obtaining 251 heads, declaring that the probability of obtaining a head upon a single toss as 0.5 presents no conceptual dilemma whatsoever.

Subjective Probability

There is yet another alternative perspective whereby $P(A)$ is taken simply as a measure of the degree of (personal) belief associated with the postulate that A will occur, the value having been assigned subjectively by the individual concerned, akin to "betting odds." Thus, for example, in rolling a perfect die, the probability of obtaining a 3 is assigned strictly on the basis of what the individual believes to be the likely "odds" of obtaining this outcome, without recourse to enumerating equally likely outcomes (the *à-priori* perspective), or

performing the die roll an infinite number of times (the *à-posteriori* perspective).

The obvious difficulty with this perspective is its subjectivity, so that outcomes that are equally likely (on an objective basis) may end up being assigned different probabilities by different individuals. Nevertheless, for those practical applications where the outcomes cannot be enumerated, and for which the experiment cannot be repeated a large number of times, the subjective allocation of probability may be the only viable option, at least *à-priori*. As we show later, it is possible to combine this initial subjective declaration with subsequent limited experimentation in order to introduce objective information contained in data in determining appropriate values of the sought probabilities objectively.

2.4 The Probabilistic Framework

Beginning with the next chapter, Part II is devoted to an axiomatic treatment of probability, including basic elements of probability theory, random variables, and probability distribution functions, within the context of a comprehensive framework for systematically analyzing random phenomena.

The central conceptual elements of this framework are: (i) a formal representation of uncertain outcomes with the random variable, X; and (ii) the mathematical characterization of this random variable by the probability distribution function (pdf), $f(x)$. How the probabilities are distributed over the entire aggregate collection of all possible outcomes, expressed in terms of the random variable, X, is contained in this pdf. The following is a procedure for problem-solving within this framework:

1. *Problem Formulation*: Define and formulate the problem appropriately. Examine the random phenomenon in question, determine the random variable(s), and assemble all available information about the underlying mechanisms.

2. *Model Development*: Identify, postulate, or develop an appropriate ideal model of the relevant random variability in the form of the probability distribution function $f(x)$.

3. *Problem Solution*: Use the model to solve the relevant problem (analysis, prediction, inference, estimation, etc.).

4. *Results Validation*: Analyze and validate the result and, if necessary, return to any of the preceding steps as appropriate.

This problem-solving approach is illustrated throughout the rest of the book, particularly in the chapters devoted to actual case studies.

2.5 Summary and Conclusions

Understanding why, despite appearances, randomly varying phenomena can be subject to analysis of any sort at all is what has occupied our attention in this chapter. Before beginning a formal discussion of random phenomena analysis itself, it was necessary to devote some time to a closer examination of several important foundational issues that are essential to a solid understanding of randomly varying phenomena and their analysis: determinism and randomness; variability and uncertainty; probability and the probabilistic framework for solving problems involving random variability. Using idealized chemical reactors as illustration, we have presented determinism and randomness as two extreme idealizations of natural phenomena. The "residence time" of a dye molecule in the hollow tube of a plug flow reactor (PFR) was used to demonstrate the ideal deterministic variable whose value is fixed and determinable precisely. At the other end of the spectrum is the length of time the dye molecule spends in a vigorously stirred vessel, the ideal continuous stirred tank reactor (CSTR). This time the variable is random and hence impossible to determine precisely *à priori*, but it is not haphazard. The mathematical model derived for the *distribution* of residence times in the CSTR—especially *how* it was obtained from first principles—provides a preview and a chemical engineering analog of what is to come in Chapters 8 and 9, where models are derived for a wide variety of randomly varying phenomena in similar fashion on the basis of underlying phenomenological mechanisms.

We also examined the characteristics of "random mass phenomena," especially highlighting the co-existence of aggregate predictability in the midst of individual irregularities. This "order-in-the-midst-of-variability" makes possible the use of probability and probability distributions to characterize ensemble behavior mathematically. The subsequent introduction of the *concept* of probability, while qualitative and informal, is nonetheless important. Among other things, it provided a non-technical setting for dealing with the potentially confusing issue of how to interpret probability. In this regard, it bears reiterating that much confusion can be avoided by remembering to keep the *concept* of probability—as a quantity between 0 and 1 used to quantify degree of uncertainty—separate from the means by which numerical values are determined for it. It is in this latter sense that the various interpretations of probability—classical, relative frequency, and subjective—are to be understood: these are all various means of determining a specific value for the probability of a specific outcome; and, depending on the situation at hand, one approach is often more appropriate than others.

Here are some of the main points of the chapter again:

- Randomness does not imply disorder; it is order of a different kind, whereby aggregate predictability co-exists with individual irregularity.

- Determinism and randomness are two extreme idealizations of naturally occurring phenomena, and both are equally subject to rigorous analysis.
- The mathematical framework to be employed in the rest of this book is based on probability, the concept of a random variable, X, and its mathematical characterization by the pdf, $f(x)$.

REVIEW QUESTIONS

1. If not a synonym for disorder, then what is randomness in scientific observation?

2. What is the concept of determinism?

3. Why are the expressions in Eqs (2.1) and (2.2) considered deterministic?

4. What is an example phenomenon that had to be ignored in order to obtain the deterministic expressions in Eq (2.1)? And what is an example phenomenon that had to be ignored in order to obtain the deterministic expressions in Eq (2.2)?

5. What are the main characteristics of "randomness" as described in Subsection 2.1.1?

6. Compare and contrast determinism and randomness as two opposite idealizations of natural phenomena.

7. Which idealized phenomenon does residence time in a plug flow reactor (PFR) represent?

8. What is the central plug flow idealization in a plug flow reactor, and how will departures from such idealization affect the residence time in the reactor?

9. Which idealized phenomenon does residence time in a continuous stirred-tank reactor (CSTR) represent?

10. On what principle is the mathematical model in Eq (2.8) based?

11. What does the expression in Eq (2.14) represent?

12. What observation by John Graunt is widely considered to be the genesis of population demographics and of modern actuarial science?

13. What are the defining characteristics of random mass phenomena?

14. How does inherent variability give rise to uncertainty?

15. What are the fundamental problems of random phenomena analysis as presented in Section 2.2.3?

16. What is the primary mathematical vehicle introduced in Section 2.3.1 for analyzing the behavior of random phenomena?

17. What is the classical (*à-priori*) perspective of probability and when is it *not* applicable?

18. What is the relative frequency (*à-posteriori*) perspective of probability and what are its limitations?

19. What is the subjective perspective of probability and under what circumstances is it the only viable option for specifying probability in practice?

20. What are the central conceptual elements of the probabilistic framework?

21. What are the four steps in the procedure for problem-solving within the probabilistic framework?

EXERCISES

Section 2.1

2.1 Solve Eq (2.11) explicitly to confirm the result in Eq (2.12).

2.2 Plot the expression in Eq (2.15) as a function of the scaled time variable, $\tilde{t} = t/\tau$; determine the percentage of dye molecules with age less than or equal to the mean residence time, τ.

2.3 Show that

$$\int_0^\infty \frac{1}{\tau}\theta e^{-\theta/\tau} d\theta = \tau \tag{2.23}$$

and hence confirm the result in Eq (2.19).

Section 2.2

2.4 The following probability distribution functions:

$$f(x) = \frac{1}{3\sqrt{2\pi}} e^{\frac{-x^2}{18}}; -\infty < x < \infty \tag{2.24}$$

and

$$f(y) = \frac{1}{\sqrt{2\pi}} e^{\frac{-y^2}{2}}; -\infty < y < \infty \tag{2.25}$$

represent how the occurrences of all the possible outcomes of the two randomly varying, continuous variables, X and Y, are distributed. Plot these two distribution

functions on the same graph. Which of these variables has a higher degree of uncertainty associated with the determination of any particular outcome. Why?

2.5 When a "fair" coin is tossed 4 times, it is postulated that the probability of obtaining x heads is given by the probability distribution function:

$$f(x) = \frac{4!}{x!(4-x)!}0.5^4 \tag{2.26}$$

Determine the probability of obtaining $x = 0, 1, 2, \ldots, 4$ heads. Intuitively, which of these outcomes would you think will be the "most likely"? Are the results of your computation consistent with your intuition?

Section 2.3

2.6 In tossing a fair coin once, describe the classical (*à-priori*), relative frequency (*à-posteriori*), and the subjective perspectives of the probability of obtaining a head.

APPLICATION PROBLEMS

2.7 For each of the following two-reactor configurations:

(a) two plug flow reactors in series where the length of reactor 1 is l_1 m, and that of reactor 2 is l_2 m, but both have the same uniform cross-sectional area $A\ m^2$;

(b) two continuous stirred tank reactors with volumes V_1 and $V_2\ m^3$; and

(c) the PFR in Fig 2.1 followed by the CSTR in Fig 2.2;

given that the flow rate through each reactor ensemble is constant at $F\ m^3/s$, obtain the residence time, θ, or the residence time distribution, $f(\theta)$, as appropriate. Make any assumption you deem appropriate about the concentration $C_1(t)$ and $C_2(t)$ in the first and second reactors, respectively.

2.8 In the summer of 1943 during World War II, a total of 365 warships were attacked by kamikaze pilots: 180 took evasive action and 60 of these were hit; the remaining 185 counterattacked, of which 62 were hit. Using a relative frequency interpretation and invoking any other assumption you deem necessary, determine the probability that any attacked warship will be hit regardless of tactical response. Also determine the probability that a warship taking evasive action will be hit and the probability that a counterattacking warship will be hit. Compare these three probabilities and discuss what this implies regarding choosing an appropriate tactical response. (A full discussion of this problem is contained in Chapter 7.)

2.9 Two American National Football League (NFL) teams, A and B, with respective "Win-Loss" records 9–6 and 12–3 after 15 weeks, are preparing to face each other in the final game—game 16—of the regular season.

(i) From a relative frequency perspective of probability, use the supplied information (and any other assumption you deem necessary) to compute the probability of team A winning any generic game, and also of team B winning any generic game.

(ii) When the two teams play each other, upon the presupposition that past record is the best indicator of a team's chances of winning a new game, determine reasonable values for $P(A)$, the probability that team A wins the game, and $P(B)$, the probability that team B wins, assuming that this game does not end up in a tie. Note that for this particular case,

$$P(A) + P(B) = 1 \tag{2.27}$$

Part II

Probability

Part II: Probability

Characterizing Random Variability

Here we have the opportunity of expounding more clearly what has already been said.

René Descartes (1596–1650)

Part II: Probability

Characterizing Random Variability

- **Chapter 3:** Fundamentals of Probability Theory
- **Chapter 4:** Random Variables and Distributions
- **Chapter 5:** Multidimensional Random Variables
- **Chapter 6:** Random Variable Transformations
- **Chapter 7:** Application Case Studies I: Probability

Chapter 3

Fundamentals of Probability Theory

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Before setting out to attack any definite problem
it behooves us first, without making any selection,
to assemble those truths that are obvious
as they present themselves to us
and afterwards, proceeding step by step,
to inquire whether any others can be deduced from these.

René Descartes (1596–1650)

The paradox of randomly varying phenomena—that the aggregate ensemble behavior of unpredictable, irregular, individual observations is stable and regular—provides a basis for developing a systematic analysis approach. Such an approach requires temporarily abandoning the futile task of predicting individual outcomes and instead focussing on characterizing the aggregate ensemble in a mathematically appropriate manner. The central element is a machinery for determining the mathematical probability of the occurrence of each outcome and for quantifying the uncertainty associated with any attempts at predicting the intrinsically unpredictable individual outcomes. How this probability machinery is assembled from a set of simple building blocks and mathematical operations is presented in this chapter, along with the basic concepts required for its subsequent use for systematic analysis of random

phenomena. This chapter is therefore devoted to introducing probability in its basic form *first*, before we begin employing it in subsequent chapters to solve problems involving random phenomena.

3.1 Building Blocks

A formal mathematical theory for studying random phenomena makes use of certain words, concepts, and terminology in a more restricted technical sense than is typically implied by common usage. We begin by providing the definitions of:

- experiments; trials; outcomes
- sample space; events

within the context of the machinery of probability theory.

1. Experiment: *Any process that generates observable information about the random phenomenon in question.*

This could be the familiar sort of "experiment" in the sciences and engineering (such as the determination of the pH of a solution, the quantification of the effectiveness of a new drug, or the determination of the effect of an additive on gasoline consumption in an automobile engine); it also includes the simple, almost artificial sort, such as tossing a coin or some dice, drawing a marble from a box, or a card from a well-shuffled deck. We will employ such simple conceptual experiments with some regularity because they are simple and easy to conceive mentally, but more importantly because they serve as useful models for many practical, more complex problems, allowing us to focus on the essentials and avoid getting bogged down with unnecessary and potentially distracting details. For example, in inspecting a manufactured lot for defective parts, so long as the result of interest is whether the selected and tested part is defective or not, the "real experiment" is well-modeled by the toss of an appropriate coin.

2. Outcome: *The result of an experiment.*

This could be as simple as an *attribute*, such as the color of a marble drawn from a box, or whether the part drawn from a manufactured lot is defective or not; it could be a *discrete* quantity such as the number of heads observed after 10 tosses of a coin, or the number of contaminants observed on a silicon wafer; it could also be a *continuous* quantity such as the temperature of reactants in a chemical reactor, or the concentration of arsenic in a water sample.

3. Trial: *A single performance of a well-defined experiment giving rise to an outcome.*

Random phenomena are characterized by the fact that each trial of the *same* experiment performed under identical conditions can potentially produce different outcomes.

Closely associated with the possible outcomes of an experiment and crucial to the development of probability theory are the concepts of the *sample space* and *events*.

4. Sample Space: *The set of all possible outcomes of an experiment.*

If the elements of this set are individual, distinct, countable entities, then the sample space is said to be *discrete*; if, on the other hand, the elements are a continuum of values, the sample space is said to be *continuous*.

5. Event: *A set of possible outcomes that share a common attribute.*

The following examples illustrate these concepts.

Example 3.1 THE BUILDING BLOCKS OF PROBABILITY
In tossing a coin 3 times and recording the number of observed heads and tails, identify the experiment, what each trial entails, the outcomes, and the sample space.

Solution:

1. Experiment: Toss a coin 3 times; record the number of observed heads (each one as an "H") and tails (each one as a "T");
2. Trial: Each trial involves 3 consecutive tosses of the coin;
3. Outcomes: Any one of the following is a possible outcome: HHH, HHT, HTH, THH, HTT, THT, TTH, TTT.
4. Sample space: The set Ω defined by

$$\Omega = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\} \quad (3.1)$$

consisting of all possible 8 outcomes, is the sample space for this experiment. This is a *discrete* sample space because there are 8 individual, distinct, and countable elements.

Example 3.2 EVENTS ASSOCIATED WITH EXAMPLE 3.1
Identify some events associated with the experiment introduced in Example 3.1.

Solution:
The set $A = \{HHT, HTH, THH\}$ consists of those outcomes involving the occurrence of *exactly* 2 heads; it therefore represents the event that "exactly 2 heads are observed" when a coin is tossed 3 times.

The set $B = \{TTT\}$ consists of the only outcome involving the occurrence of 3 tails; it therefore represents the event that "3 tails are observed."

The set $C = \{HHH, HHT, HTH, THH\}$ consists of the outcomes involving the occurrence of *at least* 2 heads; it represents the event that "at least 2 heads are observed."

Similarly, the set $D = \{HHH\}$ represents the event that "3 heads are observed."

A *simple* or *elementary* event is one that consists of one and only one outcome of the experiment, i.e., a set with only one element. Thus, in Example 3.2, set B and set D are examples of elementary events. Any other event consisting of more than one outcome is a *complex* or *compound* event. Sets A and C in Example 3.2 are compound events. (One must be careful to distinguish between the set and its elements. The set B in Example 3.2 contains one element, TTT, but the set is not the same as the element. Thus, even though the elementary event consists of a single outcome, one is not the same as the other).

Elementary events possess an important property that is crucial to the development of probability theory:

- An experiment conducted once produces one and only one outcome.
- The elementary event consists of only one outcome.
- One and only one elementary event can occur for every experimental trial.

Therefore:

Simple (elementary) events are mutually exclusive.

In Example 3.2, sets B and D represent elementary events; observe that if one occurs, the other one cannot. Compound events do not have this property. In this same example, observe that if, after a trial, the outcome is HTH (a tail sandwiched between two heads), event A has occurred (we have observed precisely 2 heads), but so has event C, which requires observing 2 or more heads. In the language of sets, the element HTH belongs to both set A and set B.

An "elementary event" therefore consists of a single outcome and cannot be decomposed into a simpler event; a "compound event," on the other hand, consists of a collection of more than one outcome and can therefore be composed from several simple events.